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Assessing Product Master Data Management Challenges in FMCG: Toward a Lifecycle Approach

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To the family I was gifted, and to the family I found.

"To the world we dream about, and the one we live in now." – Anaïs Mitchell

Abstract

This report examines the structural and procedural challenges of Product Master Data Management (MDM) within a mid-sized European healthcare distributor experiencing rapid organizational growth. I investigated the intersection of Enterprise Resource Planning (ERP) systems, data governance, and the potential role of Product Lifecycle Management (PLM) frameworks in enhancing data quality and process integration.

Employing a qualitative case study, my research maps current product data flows at an established Swedish company. I identified systemic fragmentation, inconsistent data ownership, and limited traceability across systems. The analysis reveals that the existing ERP implementation, while central to operations, lacks the flexibility and control mechanisms necessary for effective master data governance in a multi-brand, multi-channel environment.

A gap analysis highlights critical deficiencies in version control, integration protocols, and data stewardship. Two pilot interventions (standardizing product identification workflows and implementing automated change detection) demonstrate the feasibility of incremental improvements. However, the findings suggest that transitioning toward a PLM-oriented architecture is essential for long-term scalability, regulatory compliance, and cross-functional data reliability.

This study contributes to the literature on data-centric process design in distributed manufacturing and distribution networks. It offers practical recommendations for aligning ERP and PLM systems to support high-quality master data management in complex organizational settings.

Keywords PLM, product master data, data quality, product identification

Sammanfattning

Jag undersöker de strukturella och processuella utmaningarna med Product Master Data Management (MDM) inom en medelstor europeisk distributör inom hälsovårdssektorn som genomgår snabb organisatorisk tillväxt. Jag undersöker skärningspunkten mellan Enterprise Resource Planning (ERP)-system, datastyrning och den potentiella rollen för Product Lifecycle Management (PLM)-ramverk för att förbättra datakvalitet och processintegration.

Genom en kvalitativ fallstudie kartlägger jag aktuella produktdataflöden hos ett etablerat svenskt företag. Jag identifierar systemisk fragmentering, inkonsekvent dataägarskap och begränsad spårbarhet över interna och externa system. Jag visar i min analys att den befintliga ERP-implementeringen, trots att den är central för den operativa verksamheten, saknar den flexibilitet och de kontrollmekanismer som krävs för effektiv masterdatastyrning i en miljö med flera varumärken och kanaler.

Genom en gapanalys belyser jag kritiska brister inom versionshantering, integrationsprotokoll och dataförvaltning. Två pilotprojekt (standardisering av arbetsflöden för produktidentifiering och implementering av automatisk förändringsdetektering) visar genomförbarheten av stegvisa förbättringar. Resultaten tyder dock på att en övergång mot en PLM-orienterad arkitektur är avgörande för långsiktig skalbarhet, regelefterlevnad och tvärfunktionell datatillförlitlighet.

Jag bidrar genom detta arbete till litteraturen om datacentrerad processdesign i decentraliserade tillverknings- och distributionsnätverk. Denna avhandling erbjuder praktiska rekommendationer för att anpassa ERP- och PLM-system för att stödja högkvalitativ masterdatahantering i komplexa organisationer.

List of abbreviations

API	Application Programming Interface
ASIN	Amazon Standard Identification Number
BI	Business Intelligence
BOM	Bill of Materials
BtC	Business-to-Consumer
CMO	Contract Manufacturing Organization
D2C	Direct-to-Consumer
DAMA	Data Management Association
DIKW	Data-Information-Knowledge-Wisdom (pyramid)
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
EAM	Enterprise Asset Management
FBM	Fulfilment by Merchant
FBA	Fulfilment by Amazon
FMCG	Fast Moving Consumer Goods
FNSKU	Fulfilment Network Stock Keeping Unit
GDSN	Global Data Synchronization Network
GTIN	Global Trade Item Number
IM	Innovation Manager

ITB	International Trade Business
KPI	Key Performance Indicator
M&A	Mergers and Acquisitions
MDM	Master Data Management
MDOps	Master Data in Operations team
NPD	New Product Development
OTC	Over-the-Counter (pharmaceuticals)
PDM	Product Data Management
PIM	Product Information Management
PLM	Product Lifecycle Management
PMD	Product Master Data
SKU	Stock Keeping Unit

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1 Introduction

It is estimated that in 2024 around 150 zettabytes of data were created, captured, copied, and consumed worldwide, representing a compound annual growth rate of 24% (Wright, 2024). Most of this information is not saved nor retained; in a business environment, however, data storage and preservation are crucial to achieving business continuity. This information needs not only to be correctly archived, but also indexed, as proper data organisation is a requirement for easy browsing and effective value extraction (Morville, 2005).

For these reasons, data quality has become a crucial issue for all businesses working with digital data and information technology. The problems that stem from poor data quality grow proportionally to the amount of data that is stored, and the increasingly interconnected business world of today leads to widespread consequences with high impact both on costs (Gartner, 2025) and operations time (Božić *et al.*, 2024). As these symptoms gain a larger footprint on operational efficiency, companies start to act more expeditiously on data management programs.

Among these companies is FMCG Pharma AB, a consolidating enterprise specialized in marketing and distribution of cosmetic and pharmaceutical products, experiencing fast expansion across the Nordic and European market through a strategy of acquisitions of established brands. This exponential growth in people and sold goods is driving the business from a local-focused operation to an increasingly complex pan-European network of stakeholders: customers, suppliers, regulation authorities, and workforce.

From this complexity comes the natural need for progressive reorganization and optimization of processes both internal and interacting with external partners. The present dissertation arises from FPAB's needs for:

- Strengthening awareness of current processes considering recent rapid expansion
- Pushing for uniformity of ways of working, impacted by new teams joining the FPAB ecosystem as nuclear units at separate times
- Reorganizing information governance for better quality management, control, and trustability

1.1 Company introduction

FMCG Pharma AB (fictive name, from hereon, FPAB), headquartered in Stockholm, Sweden, is a pan-European consumer healthcare company that specializes in the worldwide distribution of products for personal healthcare. The product portfolio of the company includes more than 80 brands divided in four broad families: Skin Health, Foot Health, Intimate & Digestive Health, and VMS (Vitamins, Minerals, and Supplements). The workforce consists of over 450 people across 13 countries; this requires the company to rely heavily on virtual meetings and filesharing.

FPAB buys finished goods from white-label manufacturers and distributes the products through a vast network of third-party logistic partners (3PLs) and warehouses. The company engages both in business-to-consumer (BtC) market, thanks to its multiple omnichannel retailers (among which pharmacies figure as the largest partners), and in direct-to-consumer (DtC) market. DtC operations occur on both intermediating marketplaces, such as Amazon, and self-managed online web shops.

While this vast part of operations is externalised, FPAB manages in-house core business activities such as marketing and sales, demand and stock planning, and a scientific affair department that takes care of matters of quality and regulatory compliance.

The company's story starts in the 1980s to market innovative drugs originating from biotech research. In time, the company consolidated its focus on development of medicines and medical devices, also expanding in personal care and cosmetics.

After the acquisition by its current owner as of early 2025, FPAB enters a period of strong expansion through various operations of mergers and acquisitions (M&A) of major players in the Nordic and European markets. In about five years, holding on a fast evolutionary pace, FPAB turns from a local reality to a pan-European operational actor.

1.2 Problem description

As FPAB proceeds in its business development mostly through operations of M&A of other companies and brands, its product data database has been growing constantly in size and complexity.

The ERP software currently in use, IFS Cloud, was customised and introduced in 2020 to host information for an average of 300 trade items. The number of Stock Item Units (SKUs) has now increased tenfold, and the attributes stored in the software have grown in quantity, heterogeneity, and importance to the business.

As widely recognized, mergers and acquisitions are common causes for "disparate, fragmented, and inconsistent systems" (Allen and Cervo, 2015), and FPAB quickly incurred in this issue, according to management and general office judgement.

The process for M&As in the company is structured to prioritize integration of the product data most important for compliance to external requirements. Over the counter (OTC) pharmaceutical products and cosmetics are subject to stringent regulations for quality reporting in most countries (e.g. the European block). The goal for integration processes has often been limited to having a minimum viable integration rather than a full-fledged, integrated product data database, which could follow the "FPAB way" of data managing.

The "FPAB way" is a standard of recent development itself. The need for a more organized product master data database started to arise in the last couple of years, as business volumes and the total number of external partners started to grow significantly.

Currently, FPAB operates just under 80 brands, partnering with over 250 third-party partners between suppliers and 3PLs in more than 90 countries, either directly or through international trade. This results in multiple information sources and high volumes of data transfer; a situation which, without proper data governance, leads to issues in synchronisation and untrustworthiness due to the inherent complexity of data.

A company of this size, without a proper master data management strategy, may face one or more of the following risks (Allen and Cervo, 2011, ch. 1 Defining the Business Case):

- Increased costs due to operational and data redundancies or differences across lines of business
- Poorer Business Intelligence (BI) and analytics, adding to customer frustration and missed opportunities
- Customer/partner/vendor/employee dissatisfaction and consequently unrealized revenues
- Over or under delivery of services due to inconsistent product identity and tracking
- Possible overpayment of vendors and customers stemming from duplicate records
- Higher risk of audits and regulatory violations

At time of writing, FPAB has encountered some of these problems already. The following is a list of situations, as identified and further explained in Chapter 6, that exemplify the issues:

- The implementation of BI dashboards encountered multiple problems of integration and mapping between sources because of inconsistent data management;
- The monthly financial settlements often incur in mapping problems with data coming from variously built data sources and therefore create incorrect invoicing;
- Missing product data and out-of-date logistic information might limit opportunities for renegotiation of sourcing and logistic contracts.

Mark Allen and Dalton Cervo (Allen and Cervo, 2015, ch. 6) define three states to identify the level of MDM maturity in a company (Figure 1). At the beginning of this project, FPAB showed the following characteristics:

- *Multiple sources for master data*. The ERP is often referred to as a trustworthy source and the teams working closely on it actively maintain it according to this intention, but the business processes in the company at large do not often accept this assumption.
- *Disparate model*. Master data is shared between the multiple sources (ERP, planning software, file sharing folders) and synchronised in an occasional, non-standardized, unverified fashion. This manifests in inconsistencies and fragmentation.

- *Foundation for a federated model.* Despite the current issues with synchronisation, the backbone for a federated system is already set: along the years, some systems have been designed for internal two-way communication. The requirements and designed rules for processes though have not been updated according to changing business environment.
- *(Partial) data registration.* Starting 2023, team "Master Data in Operations" takes care of maintaining product data in the ERP platform and Standard Operative Procedures (SOPs) have been created to assign data governance responsibilities to different company functions. However, SOPs related to master data are not globally integrated in the company and therefore often not followed in day-to-day operations.

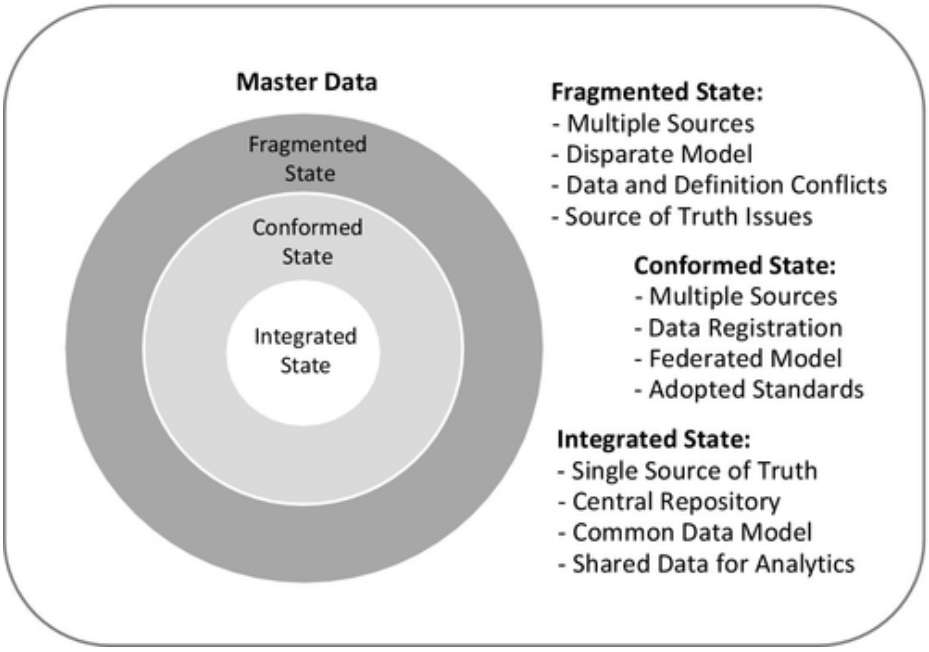


Figure 1 – Maturity levels of MDM. FPAB can be considered as in a fragmented state. Source: Allen & Cervo, 2015, Figure 6.1

FPAB can be therefore identified as applying a *fragmented state* of master data, on a path to transitioning to a conformed state; this change mainly comes from isolated initiatives of individual departments and managers. The push towards a more integrated system is a necessary first step, but what FPAB is missing is a coordinated effort with a clear mandate, common goals, and a better-defined data governance policy.

The main problems identified by management (Internal Report, February 2025) at current state are:

- Low quality of the product data database (e.g. missing, incomplete, or out-of-date data)
- Weak integration between systems, internal and external (e.g. ERP & planning software, marketplaces & 3PLs)
- Missing overarching strategy for data governance, impacting in various practical ways (e.g. undefined responsibilities for data creation and management)
- Low awareness of current data infrastructure and business needs related to data

This report aims at presenting the efforts spent to better define the impacts of these issues, to map the processes and tools involved in product data creation and management, and to propose viable solutions to improve on the gaps and weaknesses.

The main objectives commissioned to this work include:

1. Identifying the kinds of product master data in use in FPAB's processes
2. Identifying and mapping the main internal processes involving product master data and the impact of product master data quality on process success
3. Identifying and mapping the main external processes involving product master data, the actors involved in creation and update of product master data outside of company environment, and the impact of product master data quality on process success for both FPAB and the external entity
4. Identifying the software solutions involved in such processes
5. Defining requirements for future projects and/or software involving product master data to assure compliance, governance, and continuous improvement

The FPAB case is considered academically relevant because of the high organizational complexity of the company, the conspicuous number of SKUs involved, the peculiarities of the pharmaceutical and cosmetic market (the strict regulations, the mandate for traceability, the ongoing post-market surveillance). There is a notable lack of available documentation and use cases for process design and requirement analysis regarding product master data. This dissertation aims at filling this literature gap by exploring the industrial niche of a distribution business operating in the pharmaceutical and cosmetic market in Scandinavia and Europe.

2 Methodology

This dissertation was developed in close collaboration with FPAB, with a business-oriented and applied research approach. The investigated problem was jointly identified by the company and the author, following a preliminary four-month period of embedded observation within the "Master Data in Operations" team.

The aim of the research was to explore the current state of PMD management, identify inefficiencies, and propose actionable improvements. The methodology combined qualitative data collection with process mapping and early-stage implementation of selected interventions.

The study was conducted within the operational and digital infrastructure of FPAB, focusing on PMD-related processes and systems. The scope included internal workflows, external data exchanges, and the supporting IT architecture (primarily ERP and planning systems).

The primary data collection methods included:

- Participant observation: The author was embedded in the Master Data in Operations team, participating in daily activities and gaining first-hand insight into PMD workflows.
- Semi-structured interviews: Interviews were conducted with key stakeholders across departments.
- Follow-up meetings: These were used to validate findings, clarify process details, and gather feedback on proposed improvements.
- Internal documentation: Standard Operating Procedures (SOPs), ERP configuration files, integration schemas, and internal reports were reviewed to support process mapping and gap analysis.

The interview process involved 12 semi-structured organized meetings, following the scheme provided in Appendix 1. Mapping of the processes was performed using software Microsoft Visio and draw.io.

Two interventions were implemented during the research period:

1. Process reengineering of product identification workflows for e-commerce platforms
2. Automation of data quality checks for selected ERP fields, including weekly change detection and backup routines

These actions were designed based on insights from interviews and process analysis, and were evaluated through stakeholder feedback.

The research followed a pragmatic approach, prioritizing actionable outcomes and organizational relevance. While the study does not claim generalizability, it provides a detailed case-based contribution to the literature on product master data management in distributed FMCG environments.

3 Data management, MDM, and PLM

Definitions of “data” are abundant and various, depending on scope, intent, and context. Most of them, however, agree on all data sharing the following characteristics (Henderson, Earley, and Data Administration Management Association, 2017):

1. Data is a representation, directly linked to factual reality;
2. Data is meaningless without the proper context.

The rapid evolution of the FMCG sector, characterized by aggressive consolidation and digital channel expansion, has fundamentally altered the role of enterprise data: collecting it and making superficial use of it is no longer sufficient as a competitive advantage. Strong context and a direct link to factual reality need to be more important than ever in a company’s data strategy. And the way to build context for data is to have a solid master data database.

The management of Product Master Data has ceased to be a mere back-office administrative function. Instead, it has emerged as a critical determinant of operational continuity, regulatory compliance, and strategic agility. This thesis identifies a complex matrix of challenges: systemic fragmentation, inconsistent data ownership, and limited traceability, all exacerbated by a strategy of rapid M&A.

The purpose of this chapter is to construct a theoretical lens through which the specific deficiencies of FPAB’s current infrastructure (specifically the reliance on its ERP system) can be critically analysed. By synthesizing theories from information science (the DIKW hierarchy), enterprise architecture (MDM maturity models), and systems engineering (Product Lifecycle Management paradigms), this report attempts to justify the proposed transition toward a "Lifecycle Approach."

The analysis draws heavily upon the seminal works of Allen and Cervo (2015) regarding Multi-Domain Master Data Management and John Stark (2015) regarding Product Lifecycle Management. It also takes from the Data Management Book of Knowledge written by DAMA International (Data Management Association) in its 2024 Second Edition, Revised version.

Master data represents the independent and fundamental business information objects, such as customers, products, employees, suppliers, and financial structures. These provide the essential context for an organization's business transactions and analytical activities. Unlike transactional data, which records specific business events, master data objects are characterized by their relatively static nature and low volatility, serving as a trusted and authoritative source (or "single version of the truth") that must be consistently identified and interpreted across multiple organizational units and application systems. These data assets include not only unique identifiers but also extended attributes, definitions, roles, connections, and taxonomies that describe the core entities involved in operational execution.

Within this broader category, product master data specifically is defined as the comprehensive set of data defining and describing a product or service, including identifiers, classifications, and specifications required to develop, produce, and support it throughout its entire lifecycle. This domain includes critical details such as Stock Keeping Units (SKUs), quantitative measures like net weight and package size, qualitative characteristics, bills of materials (BOMs), and regulatory compliance information.

As a shared enterprise resource, product master data frames the analysis performed on transaction data and integrates information across disparate systems to resolve semantic inconsistencies and ensure high-quality information sharing across the value chain. Systems making use of product master data include Enterprise Resource Planning (ERP), Product Lifecycle Management (PLM), and Customer Relationship Management (CRM) software.

The DIKW pyramid is a traditional structural hierarchy representing the ascension from Data to Information, to Knowledge, and ultimately to Wisdom. It provides a foundational framework for understanding how organizations transmute raw symbolic representations into strategic competitive advantage.

At the base of this construct lies Data, often characterized as the raw material of information, consisting of discrete, objective facts or observations that, in isolation, lack specific meaning or intent. It must be created, captured, and managed, serving as the representation of facts concerning the enterprise's activities and resources. Consequently, product master data serves as the essential bedrock within the DIKW hierarchy, providing the reference points necessary to convert raw signals into coherent business insights.

Information is described as "data in context," implying that raw data has been processed or organized to convey meaning and relevance to a specific recipient or process. Product master data is the primary engine of this contextualization. For instance, a transactional record indicating the sale of "Item #4920" is merely data; it becomes information only when referenced against the Product Master, which identifies "Item #4920" as a specific variant of a product with defined attributes such as weight, dimensions, and colour.

Knowledge allows for the identification of relationships and clusters within the information. In the realm of product management, this is manifested through the intricate taxonomies embedded within product master data. A robust Product Master does not merely list items; it organizes them into product families, groups, and lines, thereby embedding the logic of the business into the data structure itself.

The apex of the pyramid is Wisdom, often framed in modern business contexts as "Insight". In the context of PLM, wisdom happens when the company can visualize the technical status of every product in the portfolio simultaneously, with control over authorship and revisioning. Product master data acts as the immutable spine.

Master data represents the business objects that are shared across multiple systems and processes: Products, Customers, Suppliers, and Assets. For FPAB, the "Product" domain is cen-

tral, but it is inextricably linked to the "Supplier" and "Customer" domains. Master Data Management (MDM) serves as the theoretical domain for discussing how organizations control their critical data assets.

Domain →	**PRODUCT**		**CUSTOMER & MARKET**		**SUPPLY CHAIN**	
Family →	PRODUCT MASTER DATA	PACKAGING & LABELING	CUSTOMER MASTER DATA	COMMERCIAL TERMS	SUPPLIER MASTER DATA	INVENTORY MANAGEMENT
Object →	STOCK KEEPING UNIT (SKU)	PACKAGE	CUSTOMER	PRICING AGREEMENT	SUPPLIER	INVENTORY RECORD
Attributes	SKU Code	Dimensions	Customer ID	Agreement ID	Supplier ID	Location Code
	Product Description	Weights	Name	Start Date	Name	SKU ID
	Brand	Label language	VAT Number	Discount %	Contact Information	On Hand Quantity
	Lifecycle Status	UOM (Unit of Measurement)	Country	Currency	Lead Time	Batch Number

Figure 2 - Domains of company data, with examples

Master Data Management is a comprehensive, technology-enabled business discipline that orchestrates the people, processes, and infrastructure required to ensure the uniformity, accuracy, stewardship, semantic consistency, and accountability of an organization's official shared information assets. Fundamentally, MDM seeks to decouple master data (specifically the critical business entities such as Products, Customers, and Suppliers) from the specific applications that create or fragment them. The goal is to resolve the structural inconsistencies caused by architectures behaving like silos to create a trusted "system of record" instead. Within this framework, the "Product" domain acts as a central focus that is inextricably linked to other domains; effective MDM does not merely catalogue these entities in isolation but explicitly manages the complex relationships between them, such as linking a Product to the Supplier from which it is sourced or the Customer who purchases it.

An MDM system can be built following architecture paradigms trat range in a spectrum defined by the location of data persistence, the degree of attribute consolidation, and the tightness of application coupling, primarily manifesting in Registry, Centralized, and Federated styles.

The Registry architecture represents the minimalist end of this spectrum, serving essentially as a directory that stores only the minimal set of identifying attributes necessary to distinguish between unique real-world entities. Rather than maintaining the full master record, the registry maintains pointers to the disparate "systems of record" where the rich attribute data resides. An infrastructure based on multiple, not synchronised Excel data sheets can be described as a Registry. While this approach is less invasive and faster to implement because it requires few changes to source systems, it relies on complex queries to assemble data on-demand and suffers from low consistency and coherence.

The Centralized architecture represents the maximalist end of the spectrum, where all core master data attributes are harmonized and stored within a single, authoritative repository. In this model, the centralized hub acts as the "Golden Copy," managing the entire data lifecycle (create, read, update, retire) and enforcing strict synchronization and serialization of transac-

tions. PLM implementations traditionally aim at a Centralized structure, with the PLM software acting as the only information hub. While this ensures high levels of data consistency, it can become a bottleneck and is often expensive to implement.

The Federated architecture occupies a middle ground, often utilizing a registry for identification while materializing complete master records on demand without permanent consolidation. By mapping multiple autonomous database systems into a single federated schema, this approach allows for partial and controlled sharing of data where the "master" view is assembled virtually by retrieving the best values from distributed source locations via service layers.

Enterprise Resource Planning (ERP) systems function as critical transaction-oriented applications designed to manage operative data and core business functions, such as inventory, capacity, scheduling, orders, costs, and financial revenues. Within the product domain, ERP systems typically focus on SKUs to support order entry and inventory levels, often serving as the system of record for sell-to customers or financial master data. However, despite their intention to consolidate organizational data into an information asset, ERPs often function as independent data silos that maintain their own specific data definitions.

ERP architectures are ill-equipped to handle the rigorous governance required for enterprise-wide product master data, as they prioritize the execution of specific business operations over the creation of a golden record. Furthermore, ERP systems fundamentally lack the scope to manage the complete product lifecycle: they manage the production and transaction phases but fail to encompass the full range of product definition data. Engineering designs, requirements, and specifications are managed by PDM or PLM systems, necessitating a distinct MDM environment to decouple master data from specific applications and ensure consistency. Consequently, relying on an ERP for master data storage often results in costly customization, manual data re-entry, and synchronization delays between bills of materials.

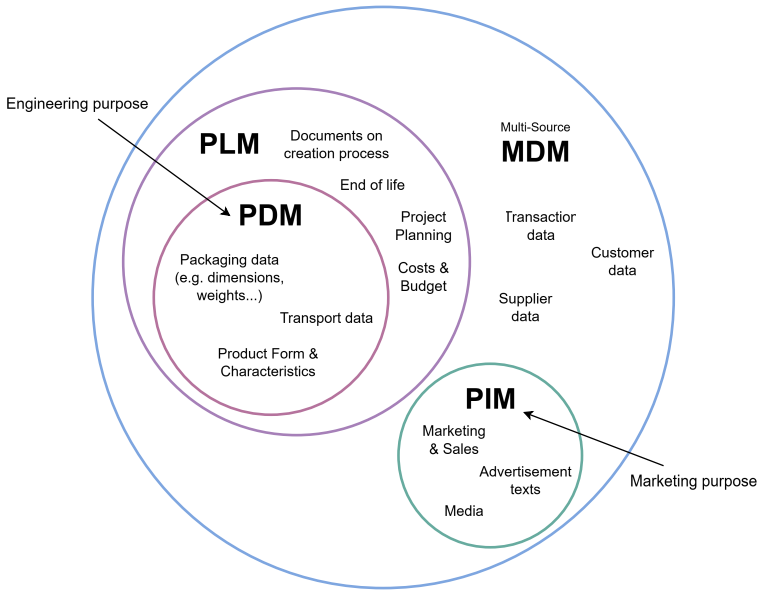


Figure 3 - Relationship between MDM, PLM, PDM & PIM

Product Lifecycle Management, when properly conceptualized, transcends its traditional characterization as a suite of software applications. Its goal is to function as a comprehensive

strategic business paradigm, designed to manage the complexity of product definition and realization throughout the entire existence of a product. Fundamental to this paradigm is the recognition that PLM is a holistic business activity that manages a company's products all the way across their lifecycles. While ERP systems are essential for the execution of business logistics and financial transactions, they are fundamentally ill-suited for the creation and evolution of product master data due to their reliance on static transactional models. The PLM paradigm acknowledges that product data is an intellectual, mobile asset and must be managed as it transitions through various lifecycle states. This process requires a system capable of managing high degrees of ambiguity and change through versions, revisions, incomplete states of drafting. These long transactions take weeks or month to complete.

PLM software adopts this iterative nature by enforcing policies fitting with the way a process behaves in the business reality: it includes the management of the data's lifecycle state (e.g., Preliminary, In Review, Released, Obsolete) and it enforces business rules that dictate, for example, that a product cannot be released to manufacturing (and thus sent to the ERP) until specific validation steps and approvals are completed. These controls stop wrong data from polluting the transactional ERP system, where such errors can cause costly manufacturing defects or supply chain disruptions. By acting as the gatekeeper of product definition, PLM ensures that the ERP system receives only high-quality, validated master data, thereby allowing the ERP to function efficiently within its own transactional paradigm without being burdened by the chaos of iteration.

Ultimately, viewing PLM as a business paradigm rather than just software reveals its critical role in the DIKW hierarchy of an organization. PLM transforms raw product data into actionable information and eventually into corporate knowledge. While ERP is indispensable for managing the flow of materials and money, PLM is indispensable for managing the definition of the assets that generate that revenue. To relegate product master data management to an ERP is to ignore the fundamental lifecycle nature of the product, whereas adopting the PLM paradigm aligns the management of data with the reality of how products are developed.

4 Current IT infrastructure

Understanding the IT systems that compose FPAB's infrastructure is key to understanding the company's processes and executing a gap analysis: in many cases, the workflows in the company are deeply shaped by the processes and limitations of the tools available.

FPAB's IT infrastructure is centralized at a company level: the same software is used and deployed in all offices, being for the most part cloud-based. For conciseness' sake, the scope of IT infrastructure analysis in this dissertation will be limited to the software that primarily manages and makes use of product master data (Figure 4). This includes the Enterprise Resource Planning (ERP) software IFS, the Sales & Operations Planning (S&OP) service optimizer software Toolsgroup's SO99+, the Microsoft 365 environment for communication and file storage, Veeva Vault for storing data and documents related to quality and compliance, Databricks and Microsoft Power BI for data analysis and visualization, and Freshworks' GLAMS for artwork lifecycle management.

IFS	ERP system
SO99+	S&OP tool
	Communication software
	File storage
databricks	Data analysis and visualization
freshworks	Artwork management

Figure 4 - IT applications in FPAB (limited to PMD use)

4.1 IFS

IFS Cloud is a Software-as-a-Service platform that FPAB adopted as its ERP software. Widely spread in the Scandinavian market, IFS Cloud is a modular platform that includes ERP, Enterprise Asset Management (EAM) and Field Service Management functionalities in a unified web app experience (IFS, 2025).

FPAB implemented the software in 2020 as a solution to increasing complexity arising in areas of Supply Chain and Finance due to an acquisition project. At the time of the acquisition the two companies were using different ERP systems: FPAB used Pyramid, the acquired company used Microsoft Dynamics. The requirements for the new platform were defined by a consulting firm together with a work group internal to FPAB.

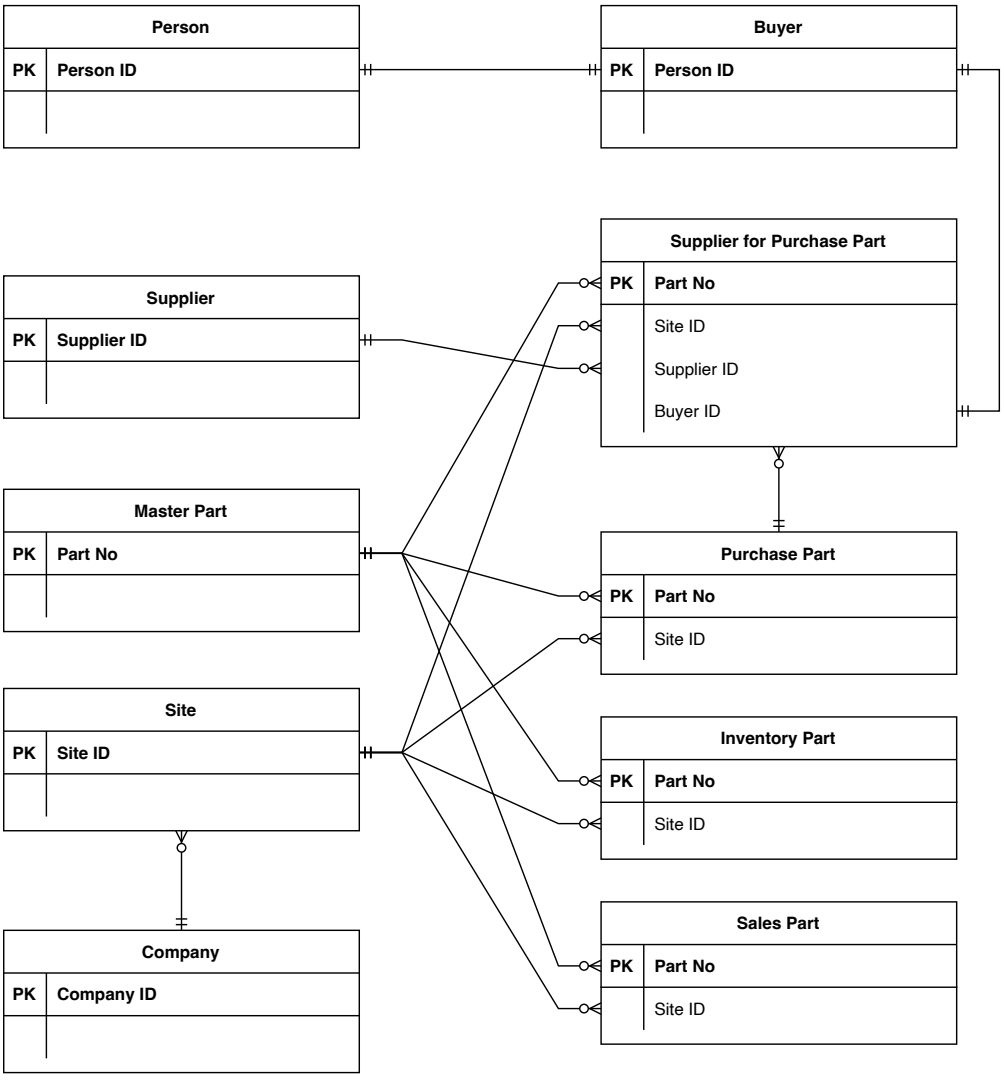


Figure 5 - Entity Relationship model of Parts in the ERP, highlighting identifiers only

The requirement analysis was conducted, according to the resulting reports, in a "very short" period, necessary for the effective integration of the newly acquired company. Furthermore, the solution was developed for a company structure extremely different from the one that can be observed today, both from a sales and a people point of view. The report estimated that the number of active users for the ERP would average at 25. Five years later, FPAB's ERP system

has more than 150 active users. The total number of article items was estimated at about 300; it is now 10 times as much.

This note is made to underline that the limitations of the current system derive, in a significant part, from the context in which it was designed. The exponential growth of FPAB in the last 5 years clearly affects the conditions under which the business operates on the ERP. While there have been occasional updates to the system's structure on occasion of the required technological upgrades, no major changes were ever made to the underlying design of the processes and data structures.

IFS is FPAB's tool for managing financial exchanges in and out of the company. The software is central to most core business processes, from sourcing to customer management. The following processes are currently implemented in IFS:

- Customer Order Management
- Procurement
- Accounting
- Master Data Management

IFS acts as a reputable source of truth for sales, stock, and product data for these processes, and this often leads to IFS being considered as a source of truth for collateral and supporting processes as well, such as artwork development or BI reporting related to stock and finance. In cases such as these, it is important to underline that not all integrations with IFS are natively supported: therefore, the chain of trustability heavily depends on the integrations between IFS and the external software.

The data model for managing product data in IFS is based on a structure divided in different "Part" entities, with each "Part" datatype being specialized in storing data for a specific business function. The full Entity Relationship model regarding product data, complete with an illustration of the Primary Keys, is schematized in **Errore. L'origine riferimento non è stata trovata.**.

Table 1 illustrates the main datatypes, their attributes, and their functionality to business processes.

Table 1 - Product Data datatypes in IFS, and relations to business processes

Datatypes	*Attributes*	*Business Process*
Master Part	The *Master Part* entity contains attributes related to: • Identifying the SKU throughout the whole company (*Part Number, Description, Status*)	Related to all business processes. The *Master Part* entity has the function of prime identifier for a SKU throughout the company, by acting as the parent entity on which all other data structures depend. The

Datatypes	*Attributes*	*Business Process*
Continues ***Master Part***	• Quantitative measures of the product aimed at identification of the trade item (*Quantity of consumer units per part, Package size, Net Weight*, etc…) • Qualitative characteristics of the product aimed at identification of the trade item (*Package type, Local marketing codes, Price visible on product,* etc…) • Codes needed for identification in external databases (*GTIN, ASIN, FNSKU,* etc…) • Package information for all levels of packaging of the trade item throughout its logistic lifecycle (weight and dimensions of the trade item, of the shipper boxes, of palletized form, etc…)	*Master Part* sets the *Part Number* and *Description* that are then propagated to all other entities and are used by the company to identify the SKU in purchase orders, customer orders, and all other documents that might be shared with external stakeholders. The *Master Part* entity can be considered as the main holder of master data information in the ERP. *Master Part* is visible to and editable from all ERP users in FPAB.
Inventory Part	The *Inventory Part* is defined as the entity describing the SKU in relation to the *Site* entity, which may represent a warehouse or a fictive site for financial purposes. The *Inventory Part* entity contains attributes related to: • Parameters necessary for demand planning (*Planning category, Launch date*, etc…) • Parameters necessary for supply planning (Lead times, *ABC Class, Shelf Life,* etc…)	Related to business processes that involve identification of physical stock in a location. Including: • Demand planning • Supply planning • Procurement • Finance *Inventory Part* is the entity that aggregates information about stock in a specific *Site*. The *Site* relationship is used as the fundamental parameter for the creation of purchase orders, invoicing, VAT calculations, and many other financial-related calculations.

Datatypes	*Attributes*	*Business Process*
Continues ***Inventory Part***	• Parameters necessary to financial management of the SKU in the specific *Site* (*Country of Origin, Custom Statistics No*, etc…) • Calculated fields summarizing important *Site*-related information (*On Hand Quantity*, etc…) • High level hierarchies for product identification (*Product, Minor Product Group*) Includes hereditary attributes from *Master Part*.	The detailed information about the stock (such as batch numbers, expiration dates, receipt dates, etc…) can be found in view *Inventory Part in Stock* accessible in the *Warehouse Management* module in IFS. *Inventory Parts* are visible to all ERP users, but editing permissions are linked to *Sites* and are user-specific.
Purchase Part	The *Purchase Part* is defined as the entity describing the procurement characteristics of a part in relation to a specific *Site*. It governs how the system interacts with suppliers and manages the replenishment of goods from external sources. The Purchase Part entity contains attributes related to: • Pricing and Purchasing Rules: Acquisition costs, currency settings, price unit of measure (UoM), and discount categories • Operational Parameters: Purchasing lead times, inspection codes, receiving tolerances, and quality control requirements	Related to business processes that involve the acquisition of materials and services from external sources. Including: • Sourcing and Procurement • Supply Chain Planning • Finance The *Purchase Part* is the entity that aggregates procurement-related rules and metadata for a part in a specific *Site*. The *Site* relationship is used as the fundamental parameter for calculating procurement lead times and defining local tax obligations required for legal compliance and financial reporting. The detailed, vendor-specific information can be found in the *Supplier for Purchase Part* view.

Datatypes	*Attributes*	*Business Process*
	• Automatic Replenishment: Settings for Purchase Requisition generation and authorization rules. Includes hereditary attributes from *Master Part*.	*Purchase Parts* are visible to all ERP users but editing permissions are site-specific.
Supplier for Purchase Part	The *Supplier for Purchase Part* is defined as the entity establishing the specific relationship between a *Purchase Part* and a *Supplier* on a given *Site*. While the *Purchase Part* defines what is being bought at a site level, this entity defines from whom and under what specific conditions the part is sourced. The *Supplier for Purchase Part* entity contains attributes related to: • Pricing and discounts: Step pricing, price breaks based on quantity, and specific currency or tax handling for that supplier • Sourcing and Supplier Management: Default supplier, manufacturer information, and supplier-specific part numbers • Logistics and Shipping: Delivery terms, ship-via codes, and specific packaging requirements.	Related to business processes that involve the management of supplier-specific commercial relationships and sourcing rules. Including: • Sourcing and Procurement • Logistics The *Supplier for Purchase Part* is the entity that defines the granular commercial agreement and logistical interface between a specific Supplier and a *Purchase Part* on a given *Site*. Detailed information regarding long-term pricing contracts and volume-based commitments can be found in the *Supplier Agreement* view, while the high-level procurement settings remain in the *Purchase Part* view. *Supplier for Purchase Part* records are visible but editing permissions are site-specific.
Sales Part	The *Sales Part* is defined as the entity representing a product or service as it is sold to a customer from a specific *Site*.	Related to business processes that involve the commercialization and distribution of products/services.

4.2 SO99+

SO99+ is an advanced supply chain planning software developed by ToolsGroup, designed to optimize demand forecasting, inventory management, and replenishment planning across complex and dynamic supply chains. It is widely adopted by mid-sized to large enterprises across industries such as pharmaceuticals, electronics, food and beverage, chemicals, and retail.

Technically, SO99+ is built around a probabilistic modelling engine that incorporates machine learning and deep learning algorithms to handle demand uncertainty and variability. Unlike traditional deterministic models that rely on single-point forecasts, SO99+ uses probability distributions to model a range of possible outcomes. This enables more accurate forecasting, especially for items with intermittent or highly variable demand such as FMCG.

The core modules include:

- Demand Planning: Uses self-adaptive algorithms to automatically model and forecast demand patterns.
- Inventory Optimization: Determines optimal inventory levels by balancing service levels with inventory immobilization.
- Replenishment Planning: Automates the replenishment process by continuously monitoring stock levels and triggering restocking notices based on real-time data.

From a business perspective, SO99+ delivers significant value by aligning inventory with customer service goals. Its service-driven approach ensures that inventory is positioned strategically across the supply chain to meet service level targets with minimal investment. This favours reduced inventory holding costs, improved service levels, and increased planner productivity.

Master data comes directly from IFS, with which SO99+ has a synchronisation pipeline set up through an external integration platform, converting the data in a format accepted by SO99+. Communication happens two-way: SO99+ also sends transactional data back to IFS in the form of purchase orders. This integration is further explored in Chapter 4.5.1.

4.3 External communications

Product master data flows to and from external partners can happen either through direct and native integration with FPAB's databases or through processes that involve human operators to export data from FPAB's environment and forward the resulting sharable output to the desired receivers through external communication methods (or vice versa, in case of incoming communication to FPAB).

Data flows with FPAB can be divided into two categories: upstream and downstream.

Upstream data flows (Figure 6) lead to new PMD creation or update in FPAB's databases. Examples of upstream data flows include communications with the CMOs regarding the finished goods and logistics information related to transportation of those goods, or communications with the 3PL partners. 3PL don't usually generate PMD, but do in case they also act as reworking hotspots for some SKUs; in those cases, a new SKU will be generated as a result of that rework and the updated PMD will be shared with FPAB. Reworks may include repackaging the pallets for specific geographic markets or relabelling consumer units to adapt for language or identification requirements for some channels and geographic markets.

The sharing activity of PMD in upstream flows is never automatic: PMD is transferred through Excel spreadsheets and email messages. Even for those partners that may have an automatic flow of transactional data (such as those 3PLs that share periodic stock status updates), the set-up of those flows is carried out entirely manually.

***Upstream** master data communication process*

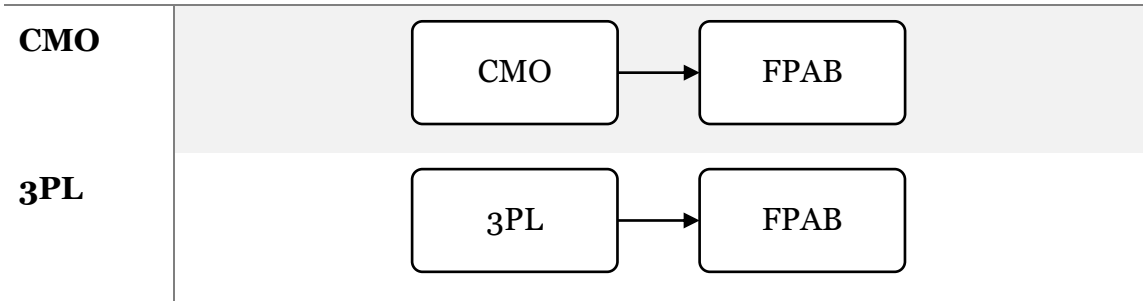


Figure 6 - Upstream communication

Downstream data flows (Figure 7) distribute FPAB's PMD to external partners. These communications happen mostly through data entry in specific online platforms or by sharing filled-in template spreadsheets via email. Different methods apply according to the commercial distribution channel. More details on downstream processes are discussed in Chapter 5.2.

FPAB operates on multiple distribution channels to sell its products:

1. Retailers
2. International Trade Business (ITB), *when business is conducted with countries where FPAB doesn't have a directly controlled local subsidiary and therefore needs to partner with an import service provider.*
3. Direct-to-consumer (D2C)

Each of these channels set their own specific technical requirements.

***Downstream** master data communication process*

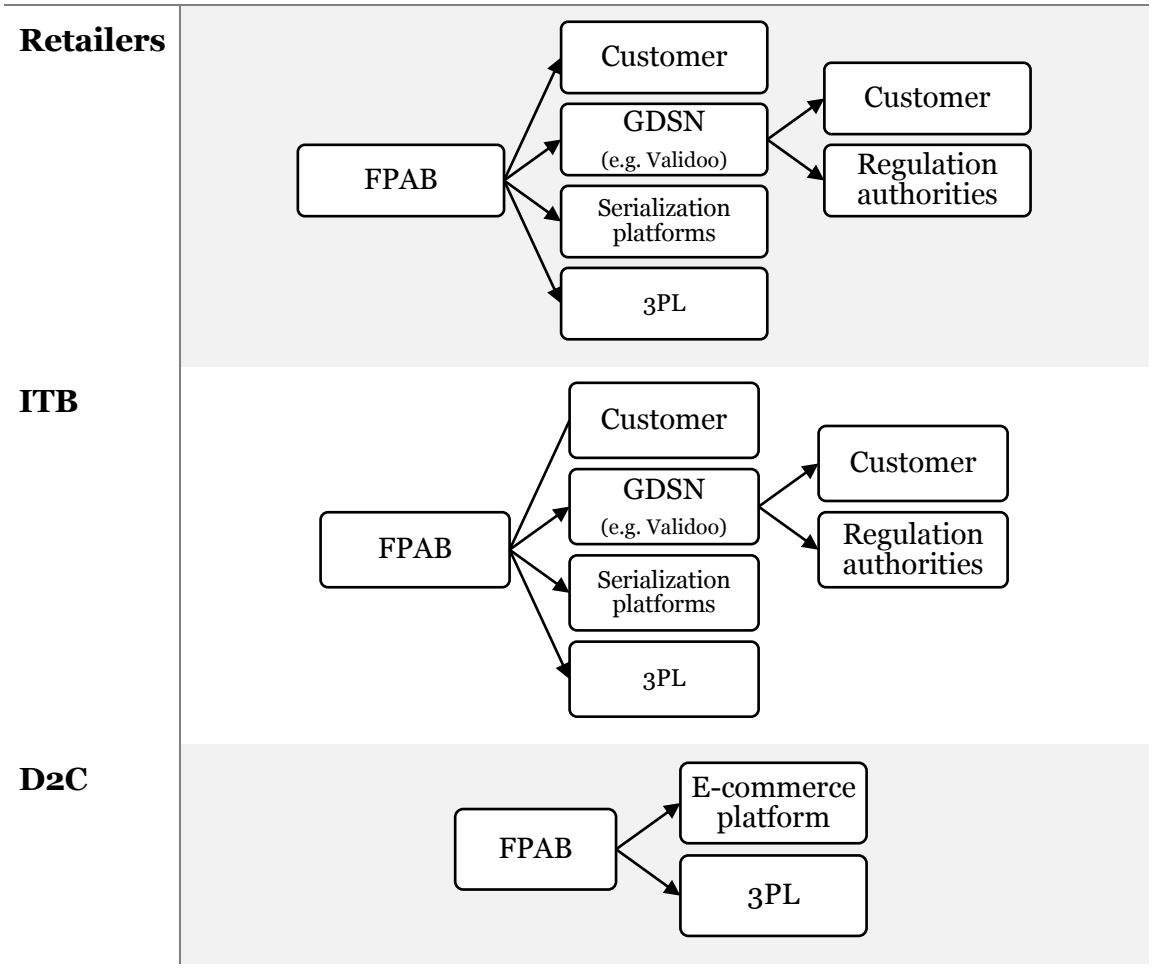


Figure 7 - Downstream communication

4.3.1 Retailers (and ITB)

For conciseness reasons, the ITB process is not explained in detail here. The process follows the same requirements and steps as the one directed to retailers, plus the possible extra step of a middle agent between FPAB and the final customer, taking care of import.

Communication of product master data to retailers happens either directly or with mediation of a data pool platform.

Direct communication happens in no case through fully automatic integration between FPAB's database and the customers'. The data flow happens either via email exchange of template files (most often, Excel sheets) or via data entry in custom online platforms setup by the customers. Preparation of these Excel files or data entry operations are often performed by key account managers in the BTC "tribe" (business-to-consumer channel vertical team). The PMD shared in these activities can be sourced from different origins, depending on how the

brand manager (or innovation manager) established the procedures for the specific product family. Among the possible sources:

- The ERP
- The product information sheet, if adopted (Excel or online spreadsheet maintained by the brand manager with master data and marketing material)
- Various files shared from CMOs and logistics partners

A previous attempt at standardizing this data collection procedure entered limited testing but was soon dropped due to a workforce change. The project involved the introduction of a Product Information Management (PIM) system synchronized directly with the ERP system. Development was halted and the platform no longer maintained, but the system is still active and future attempts at reviving the project are not excluded by the managers involved.

An alternative way to share information with the retailer customers is through centralized data pool platforms, such as Validoo for the Swedish market, which is adopted and required by many pharmacy chains.

Validoo, like many other international data pools for trade, relies on the Global Data Synchronization Network (GDSN) standard developed by GS1. GDSN is an interconnected network of communicating data pools where trading partners can upload and maintain standardized product data. This ensures that all parties have access to the latest and most accurate product information, leading to more efficient and accurate supply chains.

GDSN operates on a publish-subscribe model and is accessed by suppliers and buyers through data pools managed by national GS1 entities or by service providers that offer supplementary tools but still rely on the standardized protocol and have full access to the network. GS1 manages a Global Registry that redirects the sides involved in the transaction to the correct data pools, as illustrated in Figure 8.

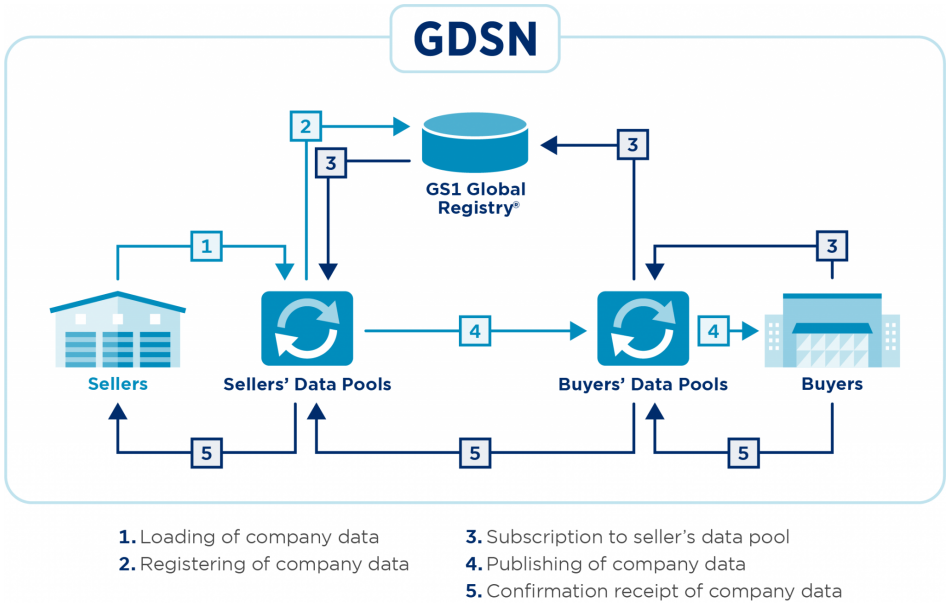


Figure 8 - GDSN network schema. Source: GS1 Sweden

4.3.2 Amazon Marketplace

Amazon is the largest online marketplace worldwide, leading among the competition in monthly visits, net and gross revenue, and market capitalization. This trend reflects in Europe as well, where the platform generates about 1270 million monthly user impressions (Web Retailer, 2023), and in the market niche of beauty and personal care. Healthcare sales on Amazon have increased 50% in revenue over the last couple years (Momentum Commerce, 2024), following a general market trend of high revenue increase through e-commerce. In the Nordics, revenue is forecasted to continuously increase between 2024 and 2029 by about 49% (Statista, 2024). Similar positive numbers apply to the European region.

FPAB successfully sells many of its products through Amazon, including a substantial set of "digital first" brands, which have e-commerce and specifically Amazon as their primary distribution channel.

Selling on Amazon can be approached with two different choices involving how to fulfil orders:

1. **Fulfilment by merchant** (**FBM**): in this scenario, Amazon will display the products on its online catalogue, handle the order process and payment from the customer, and deliver the order to the customer. However, in this case Amazon will not take care of upstream logistic operations such as warehousing the seller's inventory. When operating on "**Vendor mode**", Amazon requires a logistics network in place and will expect just the correct amount of stock necessary for fulfilling orders from the customers in a Just-In-Time approach. The logistics can be operated directly by the seller or by partnering with a 3PL.
2. **Fulfilment by Amazon (FBA)**: in this scenario, Amazon will act as a logistics partner to the seller, receiving inventory and stocking it in its warehouses. This can be considered as a complete outsourcing of the e-commerce process to Amazon: the seller will send the finished goods to the FBA network and Amazon will handle shipping, customer service, refunds, and returns. This option is commonly called "**Seller mode**".

Operating in Vendor or Seller mode has consequences on the logistic requirements that need to be applied when setting up the supply chain, including some related to PMD illustrated in Table 2. To optimize for profitability and convenience, FPAB has been operating on both modes, deciding for the best strategy on a product level.

Table 2 - Comparison of logistics requirements between fulfilment methods on Amazon

Operational area	*Fulfilment by Amazon (FBA)*	*Fulfilment by merchant (FBM)*
Inventory ownership and control	• Amazon owns stock when received at fulfilment centre • Provides visibility through periodical reports	• Seller retains full custody and control of the stock.
Trade conditions and compliance	• Handles inverse logistics • Acts as a VAT entity in some regions • Enforces internal safety and compliance standards	• Seller manages inverse logistics and stays fully responsible for all tax compliance and invoicing.
Shipping and labelling	• Has special requirements for palletisation (which may cause necessity for rework) • Has special requirements for product identification on labels for both consumer units and transport setups.	• Managed through standard logistics agreements • Necessary to relabel products according to Amazon's requirements (or incur in a fee)

4.3.2.1 Product identification

Amazon's requirements on product identification and labelling are particularly stringent and have considerable impact on how product data is shared with and by Amazon. These requirements (Table 3) vary based on whether the product is FBA or FBM (Amazon, 2025a).

The **ASIN** (Amazon Standard Identification Number) is a 10-digit alphanumerical code that is used by Amazon as a grouping identifier linked to a product entity in Amazon's catalogue. The ASIN refers, in practice, to the product page appearing on Amazon's website, uniting all offers for the same catalogue entry, independently of merchant and variations.

Table 3 - Identification requirements based on fulfilment method

*If a product is fulfilled by **Amazon**:*	*If a product is fulfilled by **merchant**:*
Every product is assigned two identification codes: an ASIN and an FNSKU	Every product is assigned one identification code: an ASIN
The primary identifier for a unit is its FNSKU	The primary identifier for a unit is its EAN/UPC barcode
The FNSKU must be present on an exterior scannable barcode that is easily accessible on every unit	The ASIN identifies the unit as part of a product family in Amazon's catalogue, but is not unique to the single unit
All other scannable barcodes must be removed, or covered, or rendered unscannable	Only one EAN/UPC barcode is allowed per unit; all other scannable barcodes must be removed, or covered, or rendered unscannable

The **FNSKU** (Fulfilment Network Stock Keeping Unit) is a 10-digit alphanumerical code that is used by Amazon as the primary identifier for units handled by the Fulfilment by Amazon program. The FNSKU uniquely identifies a trade item by its seller (allowing warehouses to distinguish between units of the same product provided by different merchants) and its eventual variations under the same ASIN.

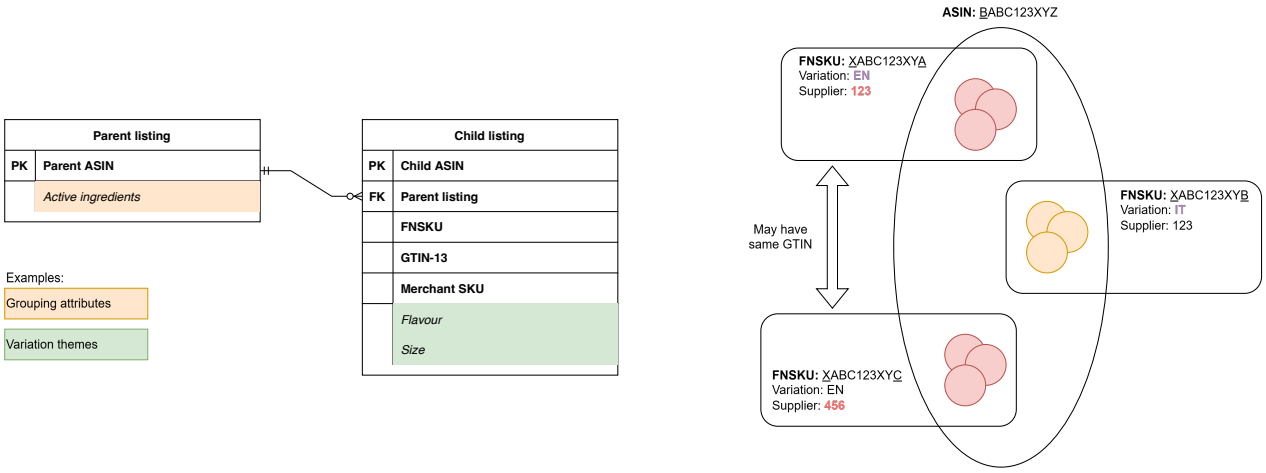


Figure 9 - Left: Entity-Relationship diagram of parent-child relationships in Amazon listings. Right: graphical representation of the relationship between ASIN and FNSKU.

Variations are, in Amazon's terminology, "sets of products that are related to one another by certain identifiers, such as size, colour or flavour". This effectively results in a parent-child relationship (Figure 9), where the parent listing is identified by a parent ASIN and is a non-buyable listing in Amazon's catalogue, while each child product is identified by its own child ASIN and an FNSKU (only if FBA). Amazon further identifies "variation themes", which are the attributes and characteristics that differentiate between child products, and "grouping attributes", which are owned by the parent listing and are identical across a variation family (Amazon, 2025b).

Each child listing will be related to multiple keys in FPAB's database, acting as identifiers in various occasions and scopes, internal and external to Amazon, as shown in Table 4. Some of these keys are mandatory from Amazon's requirements (Child ASIN, FNSKU), some are mandatory from both Amazon's and regulatory requirements (GTIN-13), and some are optional to the convenience of the merchant (Merchant SKU). Having this many identifiers for one single product opens to risk of confusion and misunderstandings during operations of data consolidation between platforms.

Table 4 - Identifier codes linked to a sellable listing on Amazon

Identifier	*Scope*	*Generated by*
(Child) ASIN	Primary key from a database perspective in Amazon's IT infrastructure	Amazon
FNSKU	Primary key used in Logistics if product orders are fulfilled by Amazon	Amazon
GTIN-13	Primary key used in Logistics if product orders are fulfilled by merchant through traditional logistics services	Merchant
Merchant SKU	Primary key used by the merchant in its internal IT infrastructure	Merchant

4.3.2.2 Product lifecycle

A product sold on Amazon follows a lifecycle composed of 2 phases:

1. Active: listing visible on the marketplace, customers can place orders
2. Inactive: listing not visible on the marketplace

4.3.2.3 Interfaces

Amazon offers the following ways to interact with its seller services:

1. Amazon Seller Central webapp
2. Amazon Selling Partner API

The Amazon Seller Central is a web application. This platform offers a web interface to create and manage Amazon product listings, accessible to merchants either via the main Seller Account (which will be the primary account administrator), via delegated Secondary Users with customizable permissions, or via Authorized Partner accounts, which are specific users created for partners external to the merchant's company such as contractors.

Amazon Seller Central offers assisted export options for transactional data from the webapp as Comma Separated Value (.csv) files. Examples of such reports include the Balance Report (for sales data) and the Inventory Report.

The Amazon Selling Partner API (SP-API) offers access to all functionalities through a REST-based API. Among the various endpoints figure methods to retrieve listings, finances, orders, outbound and inbound reports.

FPAB interacts with the platform using both the webapp (generating data exports through the webapp option) and the SP-API. The different use cases are presented in Figure 12, with further technical explanation of the integrations with IFS presented in 4.5.2.

Other actors involved in communication with Amazon's platform in FPAB's value chain are the 3PL partners. While the communication is completely internal for products fulfilled by Amazon, there is a direct channel between Amazon and the logistics partners that FPAB set up for this delivery flow. The integration is managed by the logistics partners directly and involves the use of the SP-API. Product identification is achieved through GTIN.

4.3.3 Shopify

While Amazon is the undisputed leader in e-commerce, a large majority of sellers, brands, and retailers working on the platform also sell on other marketplaces online. This diversification is important to mitigate risks, reduce dependency on one single channel, and potentially expand business to new customer audiences.

Among the most popular platforms, Shopify is chosen as a secondary marketplace by 25% of Amazon sellers (Jungle Scout, 2025, p. 10) and is the most popular e-commerce software choice in the Nordics (BuiltWith, 2025). The difference between Amazon and Shopify lies in that the latter focuses more on being a tool for D2C rather than a full-fledged generalist marketplace (Business Insider, 2021). For example, it allows for larger personalization of the store page by making the Shopify branding almost disappear behind the merchant's. On the other hand, it also does not provide any form of support in promotion.

Shopify offers no proprietary fulfilment option in Europe and UK (it does in the USA and Canada instead). Therefore, FPAB must collaborate with third-party logistics partners to keep stock in warehouses and fulfil orders. This requires a communication channel between the e-commerce platform and the logistics partners that FPAB set up for this delivery flow. The integration is mainly managed by the partners, with FPAB only acting in the setup of new products.

4.3.3.1 Product identification

Shopify identifies products and product variants through internal ID codes that are meant to be used as technical support tools only and are therefore meant to never be used in customer-facing applications. The platform then offers two fields to be optionally filled by the merchant: one is "Barcode," meant for inventory tracking by logistics partners, and the other is "SKU", meant to be filled with the merchant's internal code for tracking and reporting.

Internal standards in FPAB for filling in these two fields have never been defined. This led to non-uniform data being registered and used for product identification purposes. In some cases, the products are identified with the FPAB SKU Number, in others with the FNSKU

codes, and in others with a descriptive code built with no standard and based on the product's characteristics. This code is informally called "Norwegian Part Number" in the company, as that is the name of the field in the ERP that is (incorrectly) used to house this information. More details about the integrations between Shopify and other systems and the issues generated by non-uniform product identification can be found in Chapter 4.5.3.

Table 5 - Identifier codes linked to a sellable listing on Shopify

Identifier	*Scope*	*Generated by*
Shopify Product ID	Primary key from a database perspective in Shopify's IT infrastructure	Shopify
Shopify Variant ID	Key attributed by Shopify to product variants	Shopify
Barcode	Field used to store the key used in traditional logistics services	Merchant
SKU	See paragraphs above	Merchant

4.3.3.2 Product lifecycle

A product sold on Shopify follows a lifecycle composed of 3 phases:

1. Active: listing visible on the marketplace, customers can place orders
2. Draft: listing needs to be completed in all its part before being sold
3. Archived: listing is complete but not visible on the marketplace, as the product is no longer sold

Shopify also directly manages two exclusive status states that can be applied in the event of breach of terms and conditions.

1. Suspended: listing is hidden
2. Pending suspension: current product status (Active, Draft, or Archived) is maintained until this status is removed or changed to Suspended

4.3.3.3 Interfaces

Shopify offers the following ways to interact with its seller services:

1. Shopify Store Admin
2. Shopify GraphQL API

The Shopify Store Admin is a web application. This platform offers a web interface to create and manage Shopify product listings, accessible to merchants either via the main Admin account (which will be the primary account administrator) or via delegated secondary users with customizable permissions.

Shopify offers assisted export options for different kind of data from the webapp as .csv files. Examples of such reports include the sales and inventory reports.

4.4 Business intelligence (Power BI)

Data visualization and reporting in FPAB is, for the largest part, handled by the Finance & Digital department for dashboards of commercial, financial, and strategic interest. In a few other cases some dashboards are managed internally by other functions, such as in Operations for the dashboards focusing on KPIs related to inventory.

All data visualization is hosted on Microsoft Power BI, a cloud-based business intelligence platform. For many dashboards, Power BI covers the full data pipeline:

1. Data is extracted from various different sources through direct integrations when possible, or through data exports that are then loaded periodically into Power BI's data lake;
2. The imported data is transformed and prepared in Power BI's native tools and then loaded in the data warehouse;
3. Finally the resulting information is visualized and shared with the stakeholders through Power BI's online platform.

A recent development in the company is the progressive move to a more advanced data pipeline with the use of specialized ETL and data lake management systems.

The source of product master data for the dashboards is either IFS or SO99+ (which still gets the master data directly from IFS). The integration with IFS is managed through the native Information Source method from IFS, which is described in more technical detail in Chapter 4.5.1.

The mapping between external sources and IFS data uses, as a standard, the FPAB SKU number as a primary key (field "Part No" in IFS). However, in some cases exceptions must be made because the FPAB SKU number might not be present in the external data source.

4.5 Integration maps

4.5.1 IFS & SO99

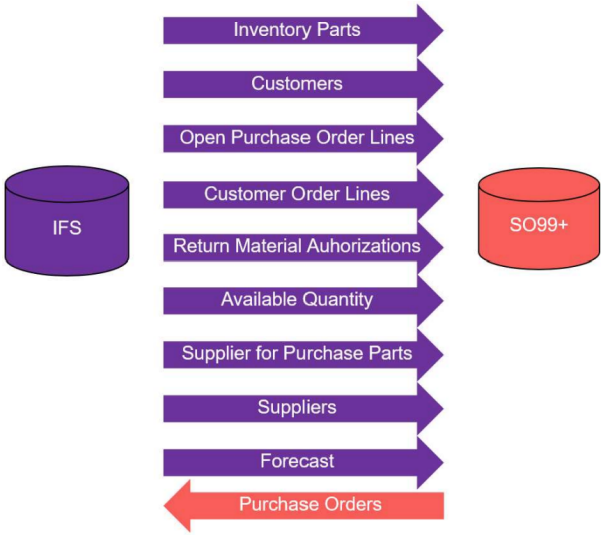


Figure 10 - Outbound and Inbound data flow between SO99+ and IFS

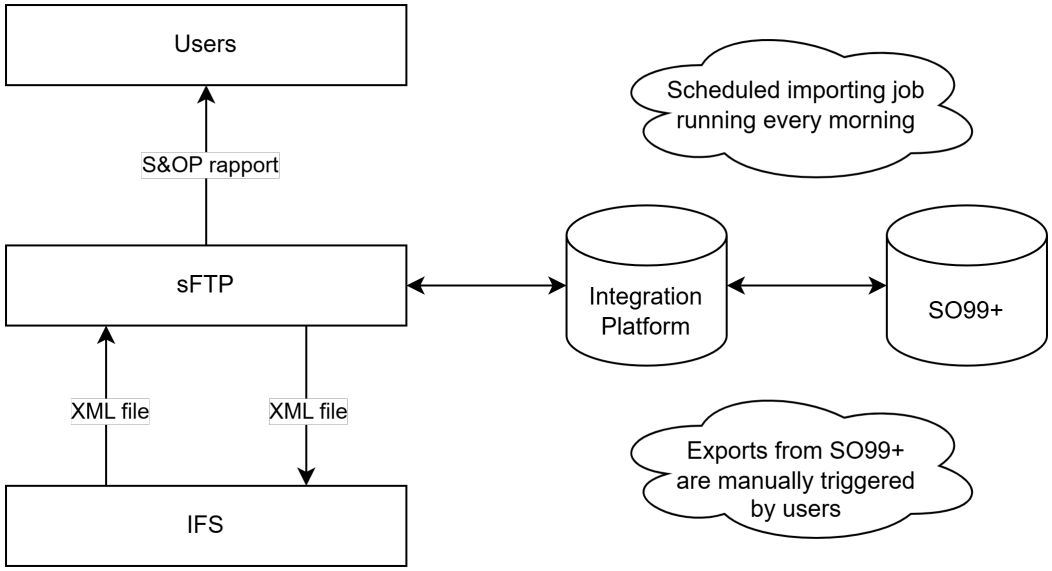


Figure 11 - Platform schema of synchronisation of SO99+ and IFS

4.5.2 IFS & web shops

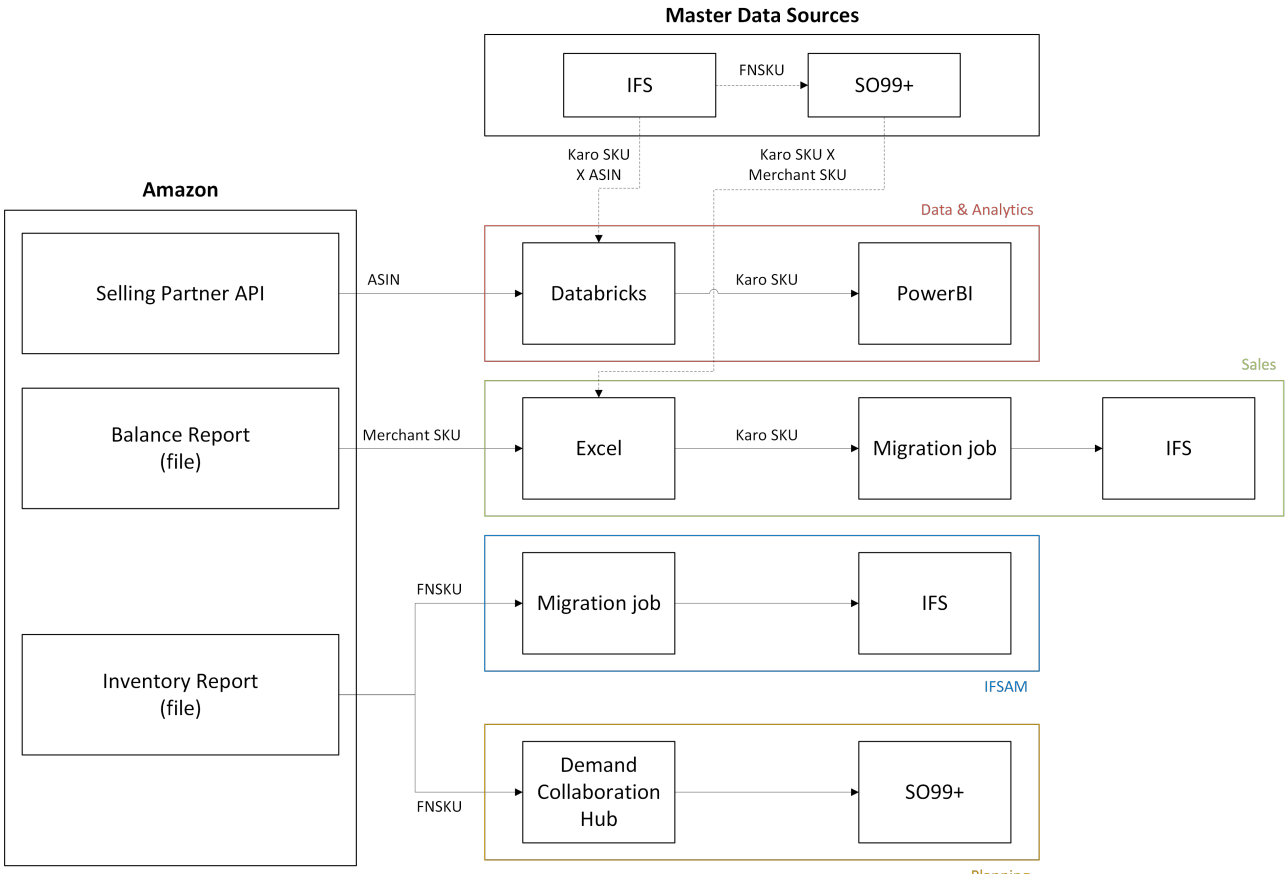


Figure 12 - Integration between Amazon and FPAB (with master data sources)

4.5.3 Web shops and 3PL

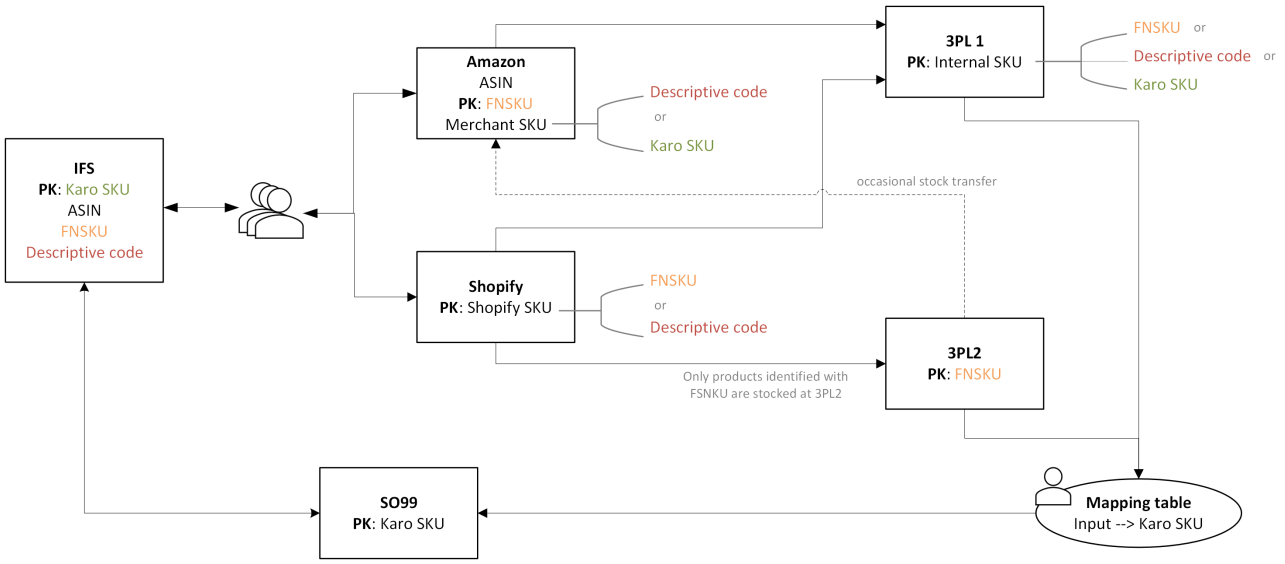


Figure 13 - Data flow between IFS, web shops, and 3PLs (focus on primary keys)

5 Current processes for Product Master Data exchange

Product Master Data (PMD) is a critical asset in FPAB's daily operations, providing a consistent and coherent foundation for all product-related activities. Within FPAB's organizational environment, the creation and distribution of PMD is a central process, taking multiple forms and involving both internal and external interactions.

This chapter will describe the status of processes for product master data exchange in FPAB. First, it will give a high-level exposition of processes dividing them between internal and external interaction processes. Then, it will highlight the main two flows of PMD between departments and partners: the New Product Development (NPD) process and a possible scenario of master data updates.

5.1 Internal PMD interactions

FPAB is internally organized into departments called "guilds," which group employees by similar professions and day-to-day activities (e.g., Operations, Finance). In addition, the company is structured into "tribes" that focus on specific distribution channel verticals, and "squads" that operate cross-functionally across brands or product categories.

Communication and data sharing practices vary across teams. While FPAB provides a suite of tools for communication, file sharing, and project management, each team is free to choose and adopt the tools that best suit their self-organizing workflows.

The current situation for PMD sharing processes between teams is of high heterogeneity. As a common point though, most teams agree on some key points that have strong impacts on the way they structure their workflows regarding PMD:

1. **The ERP is the most authoritative source in the company for product master data**

   The ERP system is widely recognized as the primary source of truth within the company, although it is not the only trusted source. In interviews, all teams acknowledged the ERP's current role as the central hub for up-to-date information on product identification and logistics configurations. While other high-trust sources are also used (see point 5), the information on the ERP is generally regarded as the preferred reference, when available.

   Consequently, most teams aim to integrate the ERP as a data source in their systems whenever feasible. However, this "feasibility" often presents a major implementation challenge, as outlined in points 2 and 3. Integration methods typically include exporting raw data from the ERP or establishing direct database connections.

Despite its central role, not all ERP data is accepted at face value. Teams often perform manual verification before relying on the information. While this practice helps prevent the propagation of potential errors, it also raises and spreads concerns about the overall reliability of the ERP dataset.

2. **Product master data quality in the ERP is often not up to business expectations (or believed not to be so)**

   PMD stored in the ERP is generally recognised as the best *available* version of the information, but also not as the best *possible* version. Years of unstructured collection of data and absent quality verification created a prejudice in the company which impacts the way processes are setup.

   Consequently, verification steps are integrated into workflows, involving not just people already in the process but also colleagues that work most closely with the desired data or are considered to be the original source for it. This leads to longer delivery times, necessary for consultations.

   Moreover, some teams created their own backups of PMD to use as an internal source of truth. These databases have often been consolidated with other sources and might correct actual errors present in the ERP, with a resulting benefit to the correctness of the process. However, this activity has problematic consequences, such as:

   - The error fixes rarely make their way back to the ERP, therefore not contributing to better data quality in the company as a whole
   - The higher trust deposited in these backups often leads to lower trust in the ERP database in cases where that data is used as a source instead
   - Syncronisation between the backup and the ERP is rarely implemented, therefore the backup fork might suffer from obsolescence
   - Errors during duplication of data in the backup can not be excluded and there are rarely verification mechanisms in place to prevent them

3. **Integration with data coming from the ERP is hard to achieve**

   It is common perception that integrating own sources and databases with data coming from the ERP is not easily achievable. While this can be true on a technical level, this widespread mindset leads to:

   - Incorrect assumptions regarding the possible options for implementation
   - Incorrect assumptions regarding the amount of support that can be provided by technical members of the team
   - Incorrect estimates of the time and work required to create such an instance of integration

   These beliefs may stop valuable opportunities for better integration between tools and datasets, reducing the general data quality in the company. What they will push teams

to do, instead, is to act as information silos either by recreating master data from scratch or by using an unsynced copy of the data as their source.

Teams that fall into such a mindset often do so due to a lack of knowledge about the tools they use, or because data quality is not prioritized in their thinking.

4. **The ERP is not an accessible software that all can use**

   The ERP is often seen by non-technical users as a tool that requires training and experience they are not equipped with. When this belief grows in team leaders, this leads to building processes in such a way that interaction with the original PMD database is not evaluated and relying on other sources of information is encouraged. When this is the team member's thought process instead, they might consider their ERP-related tasks to be of low importance; therefore, they neglect workflow steps like updating product status or characteristics when becoming aware of such an update.

5.2 External PMD interactions

Data flows between internal and external actors are visualized in Figure 14. All internal processes have been condensed in actor "Corporate functions", with the exception of "Artwork department" because of the tighter relationship between it and the external "Third party manufacturers".

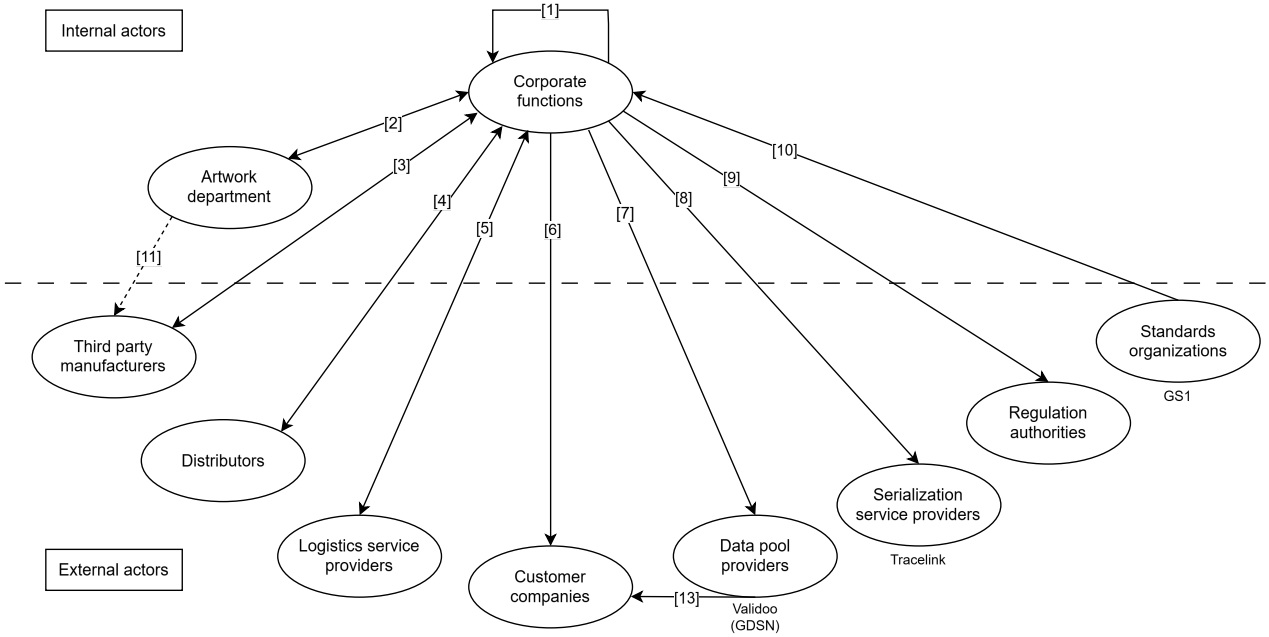


Figure 14 - PMD interactions between internal and external actors

- The master data that is generated inside the company [1] (see Figure 14) builds the main corpus of PMD that is then shared with the external actors. Corporate functions are the main and primary PMD owners and coordinators. All other data flows should center on Corporate functions as the terminal receiver of information and source of truth for their internal databases.

- Communication with the suppliers of finished goods [3] behaves as a two-way flow: the manufacturer receives product code, name, specifications for the final product, while sending back the important logistics information such as the dimensions and weight of the consumer unit. Suppliers also take care of preparing the shipping setup delivered to the 3PL. FPAB acts as an intermediary for that operation [5], receiving and validating the setup from the supplier before sharing it with the 3PL.
    - These exchanges of data happen mostly via email.
- In some rare cases, FPAB is not the actor generating and owning the GTIN codes: for specific markets, there might be a distributor or legal entity acting as a local contact and owner of the GS1 prefix in use. In those cases, the distributor will receive the necessary basic information, generate the code, share it with FPAB, and act as a point of contact for the prefix in case of need.
    - These exchanges of data happen via email.
- There is a specific flow of information happening between Corporate functions, Artwork department, and Third party manufacturers. Artwork department is an internal function, but works tightly with suppliers to coordinate on the correct printing of consumer cases and trade items. Artwork team will receive the master data and product claims from Operations and Commercial and will validate the final artwork with multiple approval stages; regarding PMD, the goal is to check identification barcodes are printed according to standard [2]. Once the artwork is approved, it is sent to the suppliers for production [11].
    - Flow [2] happens on emails, internal messagging platform, or dedicated artwork lifecycle software (the approval process is managed natively on platform). Flow [10] happens via email.
- Communication of PMD with the Logistics service providers (3PLs) moves mostly on a one-way basis [5]. FPAB shares the SKU number, description, and information of the logistic payload (e.g. dimensions, weights, quantities,…). The 3PL may use an internal identifier for the products and in some cases this code might be shared with FPAB; however, no formal process includes this code as a master data, nor includes it being stored in FPAB's systems.
    - This flow happens via email or through an online platform set up by the 3PL.

- The data flow to customers [6] is one-way: FPAB shares PMD according to the standard requested by the customer. Requirements are variable and may depend on the type of products sold: pharmaceuticals and medical devices have a higher number of attributes to be shared to adhere to national and international regulations. Communication with a larger data pool is necessary to interact with large customers such as pharmacy chains. In those cases, FPAB will upload the necessary documentation on the data pool providers [7], which will then make it accessible to final customers [13].
    - This flow uses multiple possible channels depending on the customer: Excel sheets sent by email, online webpages to be compiled, data pool providers.
- FPAB partners with specialized companies to comply with the traceability requirements for products such as pharmaceuticals and medical devices [8]. These "track & trace" services provide tools to follow the unique identifiers through the supply chain and therefore have the company be able to answer to verification inquiries from regulatory entities.
    - This flow happens via online portal provided by the external company.
- Communication with regulation authorities [9] happens both for creation and sharing of product master data information.

  National or international authorities may require for some products (pharmaceutical and cosmetics) to be registered or for a notification process to be followed before starting immission on market. Registered products will receive a registry number to be printed on consumer trade items (in some countries they also work as the primary identification key for the retailers and will therefore be used as barcode).
    - Some regulatory authorities access the product master data through public data pools [14], while others require direct communication through web portals set up by the authorities.
- GS1 is the not-for-profit organization developing and maintaining the most popular worldwide product identification standards: GTIN, barcodes, data pool communication protocols, etc… FPAB communicates with GS1 Sweden [10] to receive GTIN codes to apply on its barcodes.
    - Data generation happens on GS1's website.

5.3 Main product master data flows

5.3.1 New Product Development (NPD)

For the development of new products, FPAB follows a decision-making and operational model based on the Stage-Gate approach (Cooper, 1990). This process is led by the Innovation team, which acts as a cross-functional coordinator to all involved teams.

Currently, the process does not follow a standardized approach across all FPAB brands. This variation exists because some product families now under FPAB originated through mergers and acquisitions. In many cases, these acquisitions included not only intellectual property but also the key team members involved in those projects. As a result, certain ways of working may still reflect established habits. Moving forward, the plan is to harmonize all processes across brands under a single standardized framework. The following section outlines the most used approach today.

During the process, product master data becomes relevant between the decision gates related to the business case definition. At this stage, negotiations with the potential Contract Manufacturing Organization (CMO, or supplier) have advanced to the point where it is necessary to agree on a common method of product identification. The CMO will request a SKU number and a product description for the new item. This information will be stored in their database and will serve as the primary key for future communications between the CMO and FPAB.

The following steps represent an example case of NPD for a non-cosmetic and non-pharmaceutical product.

1. Once the CMO has been identified, the Sourcing Manager is project leader at this stage of the process. It is their responsibility to lead negotiations with the supplier and to share with them the required master data. The Sourcing Manager (SM) will request SKU number and description to team Master Data in Operations (MDOps).
2. MDOps reserves an internal FPAB SKU code, generates a description according to standard, and sometimes also reserves one or more GTIN codes for logistics purposes. The generated information is not yet entered into the ERP, but rather just saved on the Excel sheets keeping track of reserved SKU number and GTIN codes. The data is then shared with SM via email and then with the CMO.
3. The CMO will follow up by sharing the first master data information available about the product: size, net weight and gross weight are all values that get a first draft value at this stage. FPAB does not request the complete bill of materials (BOM) from its CMOs; in the cases where the BOM is shared, it is not saved in an accessible place on the ERP or shared with other teams.

   Note: the BOM will be shared with the Quality team for the purposes of compliance and due verifications. However, this process happens indipendently from the main master data flow.
4. The SM and the Innovation Project Manager will update the shared tracker used for the NPD process with the received master data. This tracker differs from brand to brand: in some cases it is shared with all actors involved in the process, in others it will stay with a smaller circle of actors. Some trackers act as a complete database of all information gathered about the product, other are used for marking step advancements only.

5. Once the business case is greenlit and the decisional gate is positive, the CMO provides the logistics information (e.g., consumer unit and shipper dimensions, weights, custom codes) and the financial master data (prices and geographical flows).
6. At this point, MDOps procedes with creating the listing on the ERP: this includes uploading both product and supply chain master data (such as components of COGS). Once the listing is active and published, it is visible to all teams in the company with access to the ERP.

5.3.2 Master Data maintenance

Master data in FPAB can change for various reasons throughout the supply chain and lifecycle of the product. Some examples include:

- Changes initiated by the supplier. These can be related to product size, product form, logistics setups, legal classifications, etc… Reason for these changes may vary from new manufacturing setups to mandatory changes made because of new regulations.
- Changes initiated by FPAB. These can be related to product SKU code, description, barcode, product size, etc… Reason for these changes may vary from internal decisions from Sourcing or Marketing to necessities for regulatory compliance. One of the most frequent causes of master data update is a change mandated by GS1 rules for updates to descriptions and EAN codes. Barcodes being updated as a consequence are among the most frequent changes.
- Changes initiated by logistics partner. These are related to logistics setups.

In all these cases, the procedure for managing changes is not standardized and there is no constant protocol to follow. The usual contact point is MDOps, who are tasked with memorizing the new information in the ERP and are expected to propagate the update to the responsible people in other functions. However, this procedure, not being codified, happens only at the discretion of MDOps and no follow-ups are coordinated for propagation and coordination. Because of the limited mapping of master data usage between software and company functions, updates are often unsynchronized between different databases.

6 Gap analysis

Previous chapters of this dissertation explored the current state of product master data flows in FPAB. This resulted in an updated mapping of the processes and IT systems involved in the creation and management of product master data along the complete lifecycle of the product, from purchase to customer delivery. The analysis touched on both internal and external data flows, focusing on the segments that are primarily FPAB's responsibilities and where external limitations condition FPAB's behaviour and ability to optimize.

This exploration was produced at a time of great change for the company, which aims at quick expansion through innovation processes and new M&As. This is a strategic moment for FPAB to set the rules and best practices that will support future business.

This chapter will perform a gap analysis between the current state of PMD exchange and a desired state of better integration, control, and quality of PMD-related workflows. The goal is to provide a clear list of weaknesses to keep under scrutiny to avoid repeating past mistakes and build toward better data health throughout all processes.

The desired state builds on the business approach of Product Lifecycle Management (Figure 15), which favours a customer-centric view on business processes and a product-focused view on technical solutions.

PLM Paradigm		*Previous Paradigm*
Business-oriented		Technically-oriented
Formally-defined		Undefined
Lifecycle		Departmental
Holistic		Piecemeal
Digital		Paper-based
Joined-up		Separate

Figure 15 - Differences between the PLM paradigm and the traditional paradigm. Source: Stark, 2015

6.1 IT infrastructure

Current state	*Desired state*	*Gap*	*Remedies*
All users with access to the ERP are part of the same user group: they share the same set of authorizations, which are currently set to Read and Write settings.	Proper data governance setup: users have Read rights on all not sensitive fields and Write rights to fields they work with only.	Missing data governance policies and criteria for assigning authorizations. Missing a set of user groups with differentiated authorizations.	Performing an exploration of user needs and a study of new user groups. Defining a contact person for authorization changes and ownership of user groups.
No version control of all PMD fields in the ERP. When any user with Write access to the field edits, not all authors or changes are memorised in the software log. No backup in the ERP to revert changes at any point in time.	All edits are monitored in all fields. All edits in all fields are listed in the integrated change log. It is possible to revert the database to its state in any recent moment in time.	No version control, no change log, no backups.	Making all PMD attributes in the ERP subject to change detection, memorizing the time and the author of the edit. Making periodic backup copies of the database (best scenario: one copy at every set of changes). Making the backups easily revertible.
Documents related to the product do not have a single library: it is spread in different positions with no index as a reference.	Allow all important documentation related to the product's lifecycle to be saved or referenced to in one single point of contact.	Missing index for finding decentralized documentation. Missing single repository for all product related documents. Missing guidelines to identify important product related documents.	Writing and publishing a guideline to identify important documents related to a product that should be easily findable and accessible. Setting up an indexing service for referencing files in a decentralized storage system; or, setting up an environment for saving and tracing all product-related documentation.
Revision versioning is available on the ERP, but functionality is not used as it would create conflicts with business requirements.	Revisions are used for keeping multiple versions of the same product active at the same time. Necessary in case of changes in the characteristics of the product and overlapping activity of two SKUs.	As the ERP is used primarily for financial purposes, having multiple versions of the same SKU at the same time is not viable.	Studying a workaround to versioning in the ERP; or, detaching PMD from software objects used for financial purposes.

Current state	*Desired state*	*Gap*	*Remedies*
PDM is stored in the ERP only and user licences are limited. This stops internal actors from accessing the data in Read-only mode.	PMD accessible to all actors involved in the product's lifecycle.	Missing platform for Read-only access to PMD for internal users.	Periodically exporting the database to a more shareable file format and location. Developing an internal visualization platform for PMD.
Inconsistency in product identification. Product primary key is not adopted in all systems in the company, or external to the company.	Single primary key used in all internal IT system. Primary key available for use in all external IT systems with periodic communication.	Missing awareness and vigilance on product identification in the IT environment.	Mandating future IT implementations in FPAB to follow standard Disseminating FPAB's primary identifier to all external databases of suppliers and 3PL (where feasible). Investigating feasibility of adding identification compliance to partners.
Low or malfunctioning integration with other IT systems in the company. Existing integrations are not maintained or verified as problems often arise. E.g., planning software.	Integration health status is verified and kept under monitoring. All internal systems communicate directly with the source of truth for PMD.	Unmonitored status of integrations between systems. Integration documentation hard to access.	Periodically verifying status of integrations. Clarifying roles and responsibilities regarding integration management. Reviewing and reworking integration documentation to encourage fixing issues without escalating.
Product families are implemented in the ERP as a label and not as an active group attribute. Every SKU entry is unlinked to other related listings, no active relationship.	Product families allow for fast identification of related SKUs and reusable/shared attributes make for easier change propagation.	Missing extensive preset product families. Missing ability to share attributes between products in the same family.	Investigate the capacities of current ERP system for applying changes to fix the gaps.

6.2 Processes

Current state	*Desired state*	*Gap*	*Remedies*
There is confusion in the company over responsibilities for updating the different fields in the ERP.	Data Governance: Each attribute has a clear owner, responsible for updating the information and reachable for requests of data updates.	Missing defined responsibilities for attributes of Product Master Data.	Leading workshops with or inside the teams to clearly define responsibilities for each attribute. Making responsibilities public to anyone either by publishing an updated SOP or by making it explicit in the ERP screens.
There is no standard or expectation over data quality stored in the ERP.	Data quality metrics are defined, public, and periodically updated and verified.	Missing data quality policy. Missing definition of data quality metrics (KPIs). Missing a public dashboard for data quality monitoring.	Writing and publishing a data quality policy to define expectations and rules regarding the state of ERP information. Leading training sessions on data quality for all users with Write access to the ERP. Defining data quality metrics for PMD attributes. Developing a publicly accessible dashboard to assess data quality level.
There is no standard or expectation on data quality stored outside of the ERP.	All employees are aware of the importance of data quality, apply policies on all systems	Missing guidelines on data quality for information exported from the ERP.	Leading training and awareness sessions on data quality for all users involved in PMD use. Writing and publishing data quality guidelines and tutorials based on company data policy.
No centralized steering or surveillance over data quality for newly setup processes involving PMD.	Data stewards verify that newly setup processes and protocol do not behave against policy and respect requirements.	No general awareness of the role data managers can fill in making processes error-proof regarding PMD.	Requiring PMD managers to verify on new SOPs involving master data creation or management. Spreading awareness of the role of PMD managers and their availability to consult on related matters.
No process for verifying changes in PMD from CMOs, pull approach only. No time schedule for regular checks.	Data in the source of truth is always updated and there are processes set to keep it so.	No process for ensuring PMD is current and up to date from external sources.	Setting up a process for recurrent checks with PMD creators (internal or external).
Inconsistency in product identification. Not all teams adopt the same standard.	Single identifier for the product allows for univocal identification of all versions.	Missing guideline and training for standard product identification.	Write guideline and conduct training for standard product identification.

7 Concluded first steps

Starting from the gap analysis as illustrated in Chapter 6, the suggested remedies were implemented for two gaps because of the reasons illustrated in Table 6. The impacts of both issues were registered in multiple teams and departments internal to the company, and the related consequences risked having an effect on the quality of interactions with operational and commercial partners. The implemented remedies have proved effective.

The two major gaps were:

Gap 1: The product identification data flow for products sold on web shops is not standardized and leads to information being lost.

Gap 2: The absence of checks for current data quality in the product master data database creates a faulty dataset; no version history on many fields also means that all data quality interventions go untracked and are therefore irreversible.

Gap 1 is process-related: the main gap lies in unstandardized operational procedures, often left to the individual judgement of the actors involved. To act on it, team Master Data in Operations built on the insights gathered from the interviews conducted for this dissertation and elaborated a reengineered process.

Gap 2 is software-related: the main gap originates from the IT infrastructure being unmatched to business requirements. Temporary solutions were pushed to the production environment to fix the current state, but a more thorough rework of the whole system will be needed to achieve a solid remedy.

Table 6 - Gaps prioritized in first actions

	Current state	*Desired state*	*Gaps*	*Remedies*
Data flow in web shop integration	Some master data information (the ASIN and FNSKU codes) are created on the Amazon platform by the FPAB team of digital marketplace managers. These codes are then sent to various departments needing it (artwork, PM, logistics) through informal and undefined methods. The communication often fails.	The master data created in the external platform Amazon is immediately transferred to the main ERP software, where all identification codes are standardized and tracked. The process for communicating this information between systems should be traceable and responsibilities should be clearly defined.	• The current process is not standardized • Communications happen with unfocused goals and methods • Responsibilities and steps to follow are not followed even when informally established	• Define actors, steps, and create documentation for a more unified and agreed on version of this process • Lead training and change management to verify implementation of the new process
Quality checks and version history for MDM dataset	No system in place to verify on the data quality of the dataset. Checks happen on individual initiative when an error is discovered in process. No version history of the dataset as a whole. Many fields have no change logging and authors of changes cannot be identified.	The master data dataset is periodically checked for consistency and general data quality. Change detection and traceability are implemented. Backup versions of the dataset are periodically created and stored.	• Software used for storing the database does not currently offer necessary options for change traceability • Authorship of changes and user authorisations are not sistematically checked and updated	• Create periodic historical backup of the master data database for reference • Introduce tool to periodically check on changes of sensitive fields with no software-integrated change detection

7.1 Processes: reworking product identification data flow for web shops

As described in Chapter 4.3, FPAB manages two main distribution channels on online web shops Amazon and Shopify. The data management and integration between these platforms and the internal FPAB system has often been difficult because of:

- unstandardized processes,
- conflicting data identification standards,
- and general low awareness regarding the goals and objectives of this process.

Both Amazon and Shopify platforms mainly use primary identification keys that are proprietary to the platform or created by the FPAB teams working on those platforms as a result of ways of working previous to the acquisition by FPAB. This resulted in databases that do not include the FPAB SKU Number that is used by most corporate functions.

Missing the FPAB SKU Number in those databases led to issues in integrating data created in those databases with the FPAB ERP environment and other tools used for business purposes. These difficulties in integration have heavily impacted on business-as-usual, as recorded in the interviews led for this dissertation and in general perception in the company. Consequences include:

- Difficulties in syncronising stock status between ERP and planning software
- Difficulties in syncronising stock status between warehouses and planning software
- Difficulties in reconciliation between the sales reports coming from Amazon/Shopify and internal reports. Further impact in this area include:
    - Additional human-hours spent in fixing inconsistent mapping
    - Generation of wrong invoicing, with effects in business risk and time spent to remedy
    - Creation of supplementary mapping table that amplify the risk for further data complexity and confusion
- Heightened risk of non-compliance to regulations
- General low trust in ERP data regarding products sold on web shops

To identify the issues causing the disruptions, a series of interviews were held with the actors that play a part in this data flow to visualize the current ways of working. The two main processes that involve product master data, as discussed in Chapter 5.3, are the NPD creation process and the regular master data management updates that happen when any attribute changes.

The **NPD** process was mapped as illustrated in Figure 16 for Amazon (only Seller mode was analysed in this instance) and Figure 17 for Shopify.

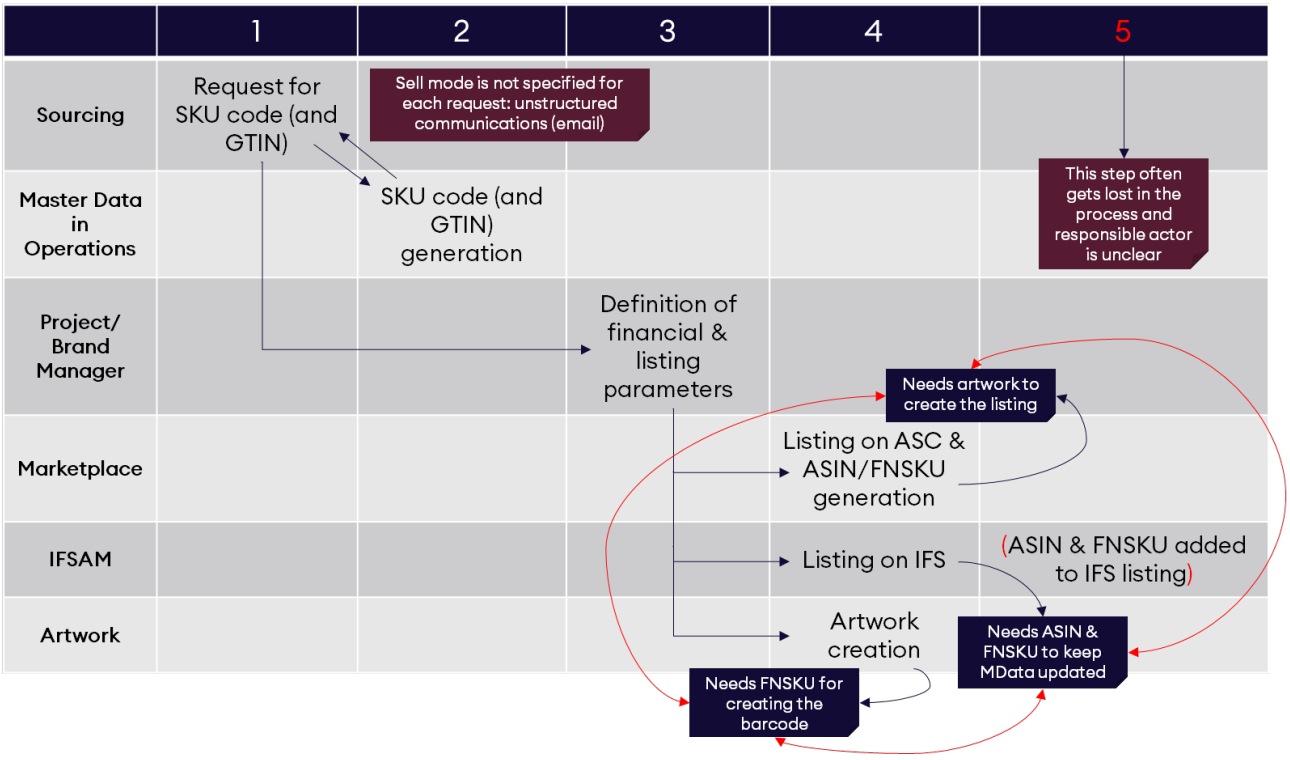


Figure 16 - AS-IS data flow in Amazon (Seller) process

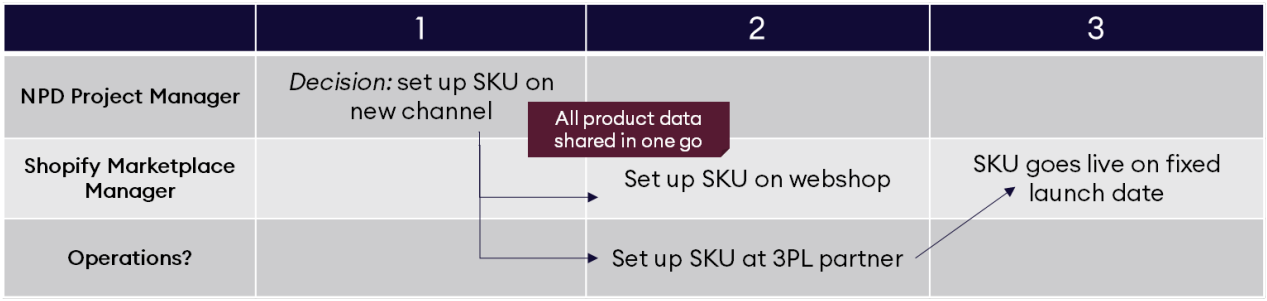


Figure 17 - AS-IS data flow in Shopify process

For the **Amazon Seller** process, the process unfolds in the following way:

0. Before the analysed data flow, the standard NPD pipeline applies. Innovation and Commercial teams plan for the product release and take care of identifying the supplier and logistic network. Then, the project is handed over to the *Sourcing Manager* and *Brand/Project Manager*.
1. The *Sourcing Manager* sends a request to team *Master Data in Operations* to generate a SKU number, description, and GTIN code for the product.
2. *Master Data in Operations* generates the requested information and sends it back to *Sourcing Manager*
3. *Sourcing Manager* hands the information to *Project/Brand Manager*, who defines the attributes of the listing for the marketplaces

At this stage in the process (steps 4 and 5), the sequence of steps becomes unclear, as multiple actions happen at the same time, generating data that is needed for progress in the other simultaneous actions.

- The *Marketplace Manager* will create the listing and generate the ASIN and FNSKU codes
    - Needs the finalized artwork to generate definitive FNSKU code
- The *Artwork* team will create the artwork
    - Needs the definitive FNSKU code to generate a finalized artwork
- The *IFS Application Management* team will create the ERP entry for the new product
    - Needs the ASIN and FNSKU codes to maintain integrations between systems

The communication between these actors does not happen in a formalized, agreed-upon manner. Exchange of information happens either through email or Microsoft Teams. Because of low awareness of the other actions happening in this process, some actors will be excluded from conversations.

The usual scenario would unfold as follows:

1. *Artwork* team produces a temporary artwork with a dummy FNSKU and barcode to pass the automatic checks imposed by Amazon before confirming the creation of the FNSKU code
2. *Marketplace Manager* receives the artwork, finalizes the listing, generates the FNSKU code and shares it with *Artwork* team
3. *Artwork* team produces the definitive artwork and send it back to *Marketplace Manager* for publishing the listing

As additional steps, the following actions might happen on personal initiative:

- ASIN and FNSKU codes are shared with the *Project/Brand Manager*; actor to perform this action is variable
- ASIN and FNSKU codes are shared with *IFSAM* team; actor to perform this action is variable

As a consequence, most often the ASIN and FNSKU codes are never entered in the ERP software, impacting all the mapping operations that regard the ERP as their central source of product master data.

After the analysed process, the product master data communications with the logistics partner are not handled by FPAB, but rather by Amazon internally).

The issue with the **Shopify** process (Figure 17) was less about complexity and more about a significant lack of synchronization. Not all the necessary stakeholders were involved: most notably, team Master Data in Operations was absent at every stage of the process. Additionally, the actions taken lacked a feedback loop to verify the accuracy of the shared information.

0. Before the analysed data flow, the standard NPD pipeline applies. Innovation and Commercial teams plan for the product release and take care of identifying the supplier and logistic network. Then, the project is handed over to the *Sourcing Manager* and *Brand/Project Manager*. As all products sold on Shopify are also sold on Amazon, the first step regarding generation of SKU number, description, and GTIN code is assumed to happen indipendently of the Shopify process.
1. If the Shopify channel is to be activated, the *Brand/Project Manager* will notify the *Shopify Marketplace Manager* that the listing needs to be created and will share the master data and marketing information via email and an Excel datasheet.
2. *Brand/Project Manager* will also interact with *Operations* team to set up the logistics partner necessary for order fulfillment. The basic master data information will be shared by the *Brand/Project Manager* via email and an Excel datasheet.

This process suffers from being unstructured in data quality and efficiency: no fixed set of master data is shared as there is no common template. The impacts of this low standardization show, for example, in one specific example of product identification that creates complications in integration between FPAB and a 3PL partner.

As shown in Chapter 4.5.3, there are cases where products from one brand are primarily identified with codes that should be proprietary and exclusive to another platform (FNSKU on Amazon). But since the barcode on the product shows that code, the FNSKU is also used in one 3PL as the barcode reference for picking operations. At the moment of partner setup, as identification guidelines were never developed in the company, the 3PL was never instructed to store the FPAB SKU number in their database and no channel of communication was ever setup to share that information. Therefore, when the 3PL shares information back to FPAB, they do so using a primary identifier that creates issues in the internal processes.

The rework process started with the Amazon Seller process. To fill in the gaps resulting from the analysis, a new process was proposed as represented in Figure 18.

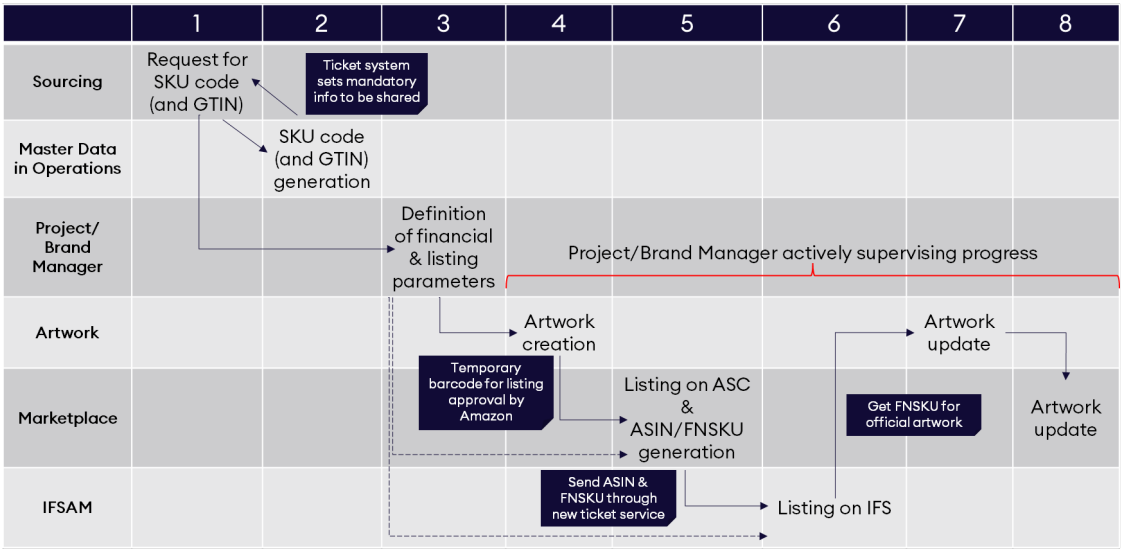


Figure 18 - TO-BE data flow in Amazon Seller process

The new process aims at being clearer on the order of actions, giving more task-specific roles, and highlighting the necessary communication loops to assure that all actors involved are updated of the evolution of the process.

The central innovation of this new process is the addition of a ticketing support service introduced as a new channel of communication between *Sourcing* and *Master Data in Operations* in steps 1 and 2 and between *Marketplace Managers* and *Master Data in Operations* in steps 5 and 6.

The benefits of having a ticketing support form in this process are:

- Providing a template and guidance on the type of master data that needs to be shared in these crucial process steps, giving more tools to validate that both data quality and completeness are respected and achieved
- Creating a formalized channel of communication where previously there was none between actros such as the Amazon Marketplace Managers (who don't have access to the ERP) and Master Data in Operations (who take care of updating product master data on the ERP)
- Providing a new tool for progress monitoring to the project manager, who will have visibility on the exchanges happening out of their specific areas of activity

The structure of the ticket form has been developed together with the actors involved and makes it mandatory to communicate important product identification codes.

This new process map and structure were presented to the stakeholders involved in May 2025 and implemented after a brief round of subsequent feedback. Change management has been supported through the creation of a manual to the revised process, a series of recorded meetings to present the new tool to the stakeholders and opening a communication channel for receiving further feedback. At time of publication, the preliminary results deliver the desired benefits regarding more effective communication and consistent recording of the identification code in the ERP system.

Rework of the Shopify process is yet to begin at time of writing. However, the plan is to integrate the ticketing system in a similar way in the immediate future. Moreover, there are plans for extensive training of relevant stakeholders (logistics and Shopify marketplace managers) in the best practices of product identification and master data sharing.

7.2 IT infrastructure: introducing automations for version history

Version control is a central feature of any PLM project: it allows to detect changes, to trace authorship and evolutions of the edits, and to easily revert to previous states of the database if needed. The current implementation of the software used for product master data storage in FPAB (the ERP, IFS Cloud) does not allow for change tracking of many fields and does not provide an easily accessible tool for version control.

Absence of an historical record for product master data led multiple times to situations of data loss or unchecked unauthorised editing of the product master data database. The impact of a similar circumstance could even lead, in the cases of some attributes, to the interruption of core operations in the company. Past incidents required many human-hours of research to rebuild the lost information and propagate all the updates, verifying that no grave consequence may have occurred in the downtime.

To close this gap from occurring again, it would be needed to upgrade the master data warehousing software to support version control and change detection in all fields. To remedy in the short term and reduce the impact of such situations, the following actions have been taken.

1. *Creation of a change detection automation on business-critical fields*

Among the attributes that do not get monitoring of edits nor authorship of the edits figure two important product identifiers: ASIN and FNSKU codes. These codes allow for mapping FPAB products to the information coming from the Amazon platform, as documented in Chapter 4.5.2.

A new routine has been setup for monitoring the evolution of ASIN and FNSKU codes in time: every Monday the Master Data in Operations team runs a data pipeline built on data analytics software KNIME to compare the current database (exported from the web interface of IFS Cloud) with the data gathered the previous week. The resulting report shows the variations on a Product and Site basis. The team will then look into the changes to verify that all the edits have been approved before being registered. If a change is unknown or unverifiable, it will be reverted to the original state.

2. *Export of a weekly backup of the Inventory Part database*

A new routine has been setup: every Monday the Master Data in Operations team downloads a dump export of the entire Inventory Parts database from the web interface of IFS Cloud and upload it in the shared folder of the Operations SharePoint server. At current state, no further analysis is performed on this dump. Future development might involve extending the change detection pipeline to more relevant fields in this database.

8 Next steps and business considerations

Building on the new mapping, the gap analysis, and the first actions already started and/or completed, a list of recommended next steps follows below. The goal with these actions would be to build remedies for the most impactful gaps identified and, as a whole, to stabilize and future-proof FPAB's processes and infrastructure. This work of consolidation and preparation would surely benefit FPAB in the long term, steadying the current state in the direction of future expansion. Five actions are recommended, plus one of more general character.

0. Start a data governance project

The first recommendation covers wider ground than just product master data, but the conspicuous lack of a data governance strategy in the company has led to notable challenges and inefficiencies that also manifested regarding PDM.

As FPAB grows in size, ungoverned decentralization of data strategies should no longer be allowed to continue. Choosing between a decentralized or a centralized structure is a multifaceted decision that should depend on the characteristics of the organization's structure and its objectives for the future (Weber, Otto and Österle, 2009). However, even decentralization necessitates of governance— possibly even more than a centralized approach.

Data governance strategies benefit the company in a variety of ways, direct and indirect. But also, master data management benefits data governance as a bottom-up approach: structured MDM projects support data governance by consolidating the foundation of the whole data environment, defining data quality in measurable effects, and by reestablishing roles and responsibilities (Hikmawati, Santosa and Hidayah, 2021).

1. Move forward in a feasibility study for implementation of PLM software

The evidence collected in this dissertation proves that the current situation of product MDM—storing the product master data in a customized module of the ERP—is not sufficient to FPAB's needs anymore and severely limits quality and process efficiency.

Some limitations could be possibly patched by modification to the current software (e.g., better authorisation profiles or adding edits of some fields to the change log), but deeper flaws are intrinsic to the nature of an ERP. For example, the consequences of using the same object as a master source for operational and financial processes (e.g. having planning status linked to innovation/sourcing status) or the absence of functionalities traditionally advantageous in PLM implementations (e.g. document version control). Furthermore, a PLM software could open the option to future improvement opportunities in other processes, such as sourcing and sustainability (see below).

It is recommended that, building on this document defining needs and requirements, a more pragmatic exploration is started regarding the actual technical and economical feasibility of an implementation project.

2. **Start training efforts for raising awareness on importance of product master data quality in FPAB**

While FPAB is currently limited by its IT infrastructure in its efforts to make MDM more efficient, research shows that "for larger organizations, issues related to IT systems seem to be much less of a problem than organizational issues" (Vilminko-Heikkinen and Pekkola, 2017). This can also be the case for FPAB: having no data governance guidance throughout the M&As possibly led the company's master data protocols to the disjointed state they are in now.

To ensure success of future attempts to reorganize MDM, it is recommended that FPAB takes action to educate the people working with master data about the tools, the processes, and the related quality protocols. From the interviews led for this dissertation, it is clear that the general perception is that the available materials and knowledge are not enough to accommodate for FPAB's needs. This applies not just to master data specific tools, but in general about the software most used in the company (IFS and SO99) which users show low confidence in their use.

3. **Review all integration protocols between software, internal and external**

Current integration flows are not all currently working as expected when introduced. One example to this behaviour is the approach to product identification. Many integrations with the ERP were set up to synchronize based on the FPAB SKU number and, in some cases, to use the ERP database as a mapping reference with different codes. As data quality in the ERP suffered, teams started to integrate custom static mapping tools in their systems, which now may have obsolete and never updated information. It is recommended to verify that all integrations are currently working according to specification.

4. **Define and monitor product master data quality metrics**

During the past two years, team Master Data in Operations led an effort to significantly improve the quality state of the PMD database in the ERP. This project involved reviewing the information already present and contacting the CMOs and 3PL partners to verify it or fill in the gaps. The activity can be considered successful based on anecdotal evidence, but no measurement system is in place to assess that.

It is recommended that, together with the data governance steering committee, a set of KPIs is defined to monitor the quality level of the database, setting goals for the short and long term. This will support teams in cases where a new software is introduced, new integrations are developed, or a new dataset is to be implemented in the database.

5. **Define minimum acceptable level of database and system integration for future M&As**

Looking ahead, further M&As remain well within FPAB's strategic scope. To support the integration of PMD across entities, it is recommended to define clear quality requirements and guidelines. This will help prevent a recurrence of issues where poor PMD quality undermines operational processes and erodes trust in the database. These requirements should align with

the established quality KPIs for the broader database. Additionally, a contingency plan with corrective actions should be in place to address any decline in data quality below acceptable thresholds.

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Some possible future opportunities have also been identified in relation to this project and its goals:

1. Resume developing of the PIM software project. A PIM software would help with standardization of product information shared with customers, including but not limited to master data. PIM software allow to centralize management of marketing claims and material and generating product data sheets faster thanks to templates and customizable presets.
2. Explore feasibility and value of storing more detailed product data (packaging, BOM, sustainability). The same PLM software can be of use to multiple teams in the company for different purposes. Leading workshops to assess common needs and opportunities could reveal ways to optimize on economical and operational ways to implement the project.
3. Investigate if sharing PMD with customers through common data pools (such as other GS1-powered platforms as Validoo) may reduce the amount of time and errors spent on the activity.

9 Conclusion

This project has examined the challenges and improvement opportunities in Product Master Data Management at FMCG Pharma AB (fictive name), a mid-sized FMCG company operating in a highly regulated and rapidly evolving market. Through a qualitative case study approach, the research mapped current PMD processes, identified systemic and organizational gaps, and implemented targeted interventions to address critical weaknesses.

The findings reveal that while the ERP system serves as the central repository for product data, it lacks the flexibility, traceability, and lifecycle orientation required to support high-quality master data management in a complex, multi-brand environment. The absence of version control, inconsistent data ownership, and fragmented integration practices have contributed to widespread inefficiencies and data quality issues.

Two pilot interventions (standardizing product identification workflows and introducing automated change detection) demonstrated the potential for low-cost, high-impact improvements. However, the analysis also underscores the need for a more fundamental transformation, including the adoption of a PLM system and the establishment of a formal data governance framework.

The study contributes to both academic and practical understanding of MDM in post-M&A contexts, particularly within the pharmaceutical and cosmetic sectors. It highlights the importance of aligning technical infrastructure with organizational processes and emphasizes that sustainable data quality improvements require not only tools but also cultural and structural change.

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Appendix 1: Interview scheme

- *Give a definition of product master data to start from common ground*
- How does this definition differ from your personal knowledge?
- In which occasions would you say you interact with product master data in your work?
- How does the product MDM process currently proceed in your opinion?
    - What is the role of your team in the process?
    - What is your role in the process?
    - What data do you receive from other teams?
    - What data are you giving to other teams?
- Does the current process / do the current operations function properly in your opinion?
    - Why / Why not?
- Which tools/software do you access PMD from?
    - Which issues can you relate to the software you use? Why?
- What data quality issues have you encountered, and how were they resolved?
- Does your team have any internal data governance/quality policies you apply?
- What triggers the creation or update of product master data?
- How do you manage historical data?
- Do you feel like you would need more support or training on the topic?

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